Fear and Safety Learning in Anxiety- and Stress-Related Disorders: An Updated Meta-Analysis

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Abstract

Fear learning processes are believed to play a crucial role in the development and maintenance of anxiety and stress-related disorders. To integrate results across different studies, we conducted a systematic meta-analysis following PRISMA guidelines to examine differences in fear conditioning during fear acquisition, extinction, and extinction recall between individuals with anxiety-related or stress-related disorders and healthy participants. This analysis updates the work of Duits et al. (2015) while also refining distinctions between physiological and behavioral outcomes and examining extinction recall. Our meta-analysis encompasses 77 studies published from 1986 to 2022, involving 2052 patients with anxiety disorders, OCD, or PTSD, and 3258 healthy controls. The results indicate significant differences in fear acquisition, extinction, and recall between the two groups. Specifically, during acquisition patients exhibited heightened physiological and behavioral responses to the CS- and reported increased affect ratings for the CS+. During extinction and extinction recall, patients continue to show heightened threat expectancy and negative affect ratings towards the CS- and increased affect ratings towards the CS+. No differences were found in CS+/CS- differentiation between groups. These findings imply that individuals with anxiety and stress-related disorders may exhibit amplified responses to safety cues and stronger reactions to threat cues during fear conditioning, lasting through extinction and extinction recall. These changes may lead to increased sensitivity in detecting fear and slower extinction process, resulting in more enduring anxiety responses. We discuss these results in the context of existing literature on fear and safety learning and consider potential underlying mechanisms.

Keywords: *Fear conditioning, Anxiety- and stress-related disorders, Fear acquisition, Fear extinction, Psychophysiological measures, Startle response, Skin conductance response (SCR), Updated meta-analysis*

1. Introduction

Acquiring the ability to recognize threats, adapt behavior in response to potential dangers, and protect oneself from harm are fundamental characteristics of human behavior that can be crucial for survival. However, excessive fear and anxiety can lead to behavioral impairments and distress, as seen in individuals affected by anxiety-, stress-related disorders, and obsessive compulsive disorder (OCD; American Psychiatric Association, 2013), which are the most prevalent mental health disorders globally (Kessler et al., 2005). Anxiety disorders impose a significant burden on individuals, compromising their overall quality of life, and also carry substantial socioeconomic costs (Konnopka & Konig, 2020; Rehm & Shield, 2019). Alterations in fear and safety learning processes have been identified as crucial in the onset and maintenance of anxiety- and stress-related disorders (Craske et al., 2022; Duits et al., 2015). Exposure-based therapies are an evidence-based and highly effective form of cognitive-behavioral therapy (CBT)[1] for anxiety, post-traumatic stress disorder (PTSD) and OCD, which specifically target the underlying learning processes (Hofmann & Smits, 2008). However, a significant proportion of individuals do not respond to these treatments, and symptom recurrence is a common phenomenon (Craske & Mystkowski, 2006; Pittig et al., 2021; Springer et al., 2018). Therefore, advancing our understanding of the mechanisms contributing to anxiety and the persistence of anxiety symptoms is crucial for refining existing treatment modalities and developing innovative interventions (Craske et al., 2022).

In laboratory settings, Pavlovian fear conditioning paradigms have emerged as a central tool for investigating fear learning processes, resulting in significant advancements of our basic understanding of the underlying mechanisms, and further enabling researchers to develop and refine effective treatment options (Beckers et al., 2023; Craske et al., 2022; Craske et al., 2014; Vervliet & Boddez, 2020). The core process in fear conditioning involves presenting an initially neutral stimulus (NS), paired with an inherently aversive or threatening stimulus (unconditioned stimulus; US). Through multiple repetitions of this pairing, an association is established between the NS and the US, known as fear acquisition. During fear acquisition, the NS becomes a conditioned fear stimulus (CS+) that can elicit conditioned fear responses in the absence of the US, termed

[1] Abbreviations: ACC = Anterior Cingulate Cortex; CBT = Cognitive-behavioral therapy; CR = Conditioned response; CS+ = Conditioned fear stimulus; CS- = Safety-associated stimulus; EMA = Ecological momentary assessment; ESA = Experience sampling method; HC = Healthy controls; EEG = Electroencephalogram; fMRI = Functional Magnetic Resonance Imaging; FPS = Fear potentiated startle; GAD = Generalized anxiety disorder; MIX = patient sample with various anxiety disorders; NS = Neutral stimulus; OCD = Obsessive-compulsive disorder; PD = Panic disorder; PFC = Prefrontal cortex; PTSD = Post-traumatic stress disorder; RSA = Representational similarity analysis; SAD = Social anxiety disorder; SCR = Skin conductance response; SMD = Standardized between-group mean difference; SP = Specific phobia; US = Unconditioned stimulus; vmPFC = Ventromedial prefrontal cortex

conditioned fear responses (CR; Pavlov, 1927). In the context of differential fear conditioning, a second neutral stimulus is introduced during fear acquisition but is never paired with the US, thereby becoming a safety-associated stimulus (CS-). In addition to the responding to CS+ and CS-, their differentiation, e.g. the difference in response to the CS+ versus the CS-, is a commonly used outcome in fear conditioning studies (Lonsdorf et al., 2017). During extinction learning, the CS+ is presented repeatedly without the aversive US (now CS+/noUS), which leads to a decrease in the CR (Hermans et al., 2006; Milad & Quirk, 2012). More specifically, during fear extinction a new stimulus-outcome association for the CS+, i.e. a safety representation for the absence of the aversive outcome is formed. Studies support the assumption that the original memory trace (CS+/US) is not erased but coexists with the new learning trace (CS+/noUS) but is inhibited by the newly formed association (Kim & Jung, 2006; Myers & Davis, 2007). In addition to the most frequently examined processes of fear acquisition and extinction, some studies also investigate the persistence of extinction learning by incorporating an extinction recall phase. This phase typically occurs either immediately or within a delay, often 24 hours or more, after extinction training and does not involve any active experimental manipulation. During extinction recall, the CS+/noUS is presented again to test whether the extinction memory is retrieved and expressed (Lonsdorf et al., 2017). The purpose of this phase is to evaluate whether extinction retention occurs, indicating either the maintenance of extinction or a spontaneous renewal or recovery of the conditioned fear response (CR). Essentially, it assesses whether the fear memory or the extinction memory trace is dominant. This phase is particularly interesting from a clinical perspective, considering the frequent reports of the recurrence of anxiety, OCD, or PTSD symptoms after successful exposure therapy (Levy et al., 2021).

The formation of associations and the learning process, i.e., whether fear and safety learning have occurred, can be investigated at various levels, including verbal reports on affective experiences, cognitive evaluation, physiological responses, and neurobiological mechanisms (Lonsdorf et al., 2017). According to the dual-process account of fear learning (Hamm & Weike, 2005; LeDoux & Pine, 2016; Sevenster et al., 2012), distinct learning processes are at work that can be assessed through different outcome measures. More specifically, this framework posits the existence of two emotional learning systems in fear responses: a rapid, automatic system responsible for reflexive fear reactions, such as physiological responses, and a slower, controlled system involved in cognitive fear responses that consider contextual information and previous experiences, as measured through ratings. This highlights the significance of incorporating diverse measurements and units of analysis for a comprehensive understanding of fear and safety learning. Verbal reports of the affective valence of the CS+ and CS- as well as cognitive evaluations of threat

expectancy occurrence are among the most common outcome measures in the fear conditioning domain (Lonsdorf et al., 2017). On a psychophysiological level, skin conductance responses (SCRs) and fear potentiated startle (FPS) represent commonly utilized indices of fear conditioning (Lonsdorf et al., 2017). Both represent bodily responses mediated by the autonomic nervous system (Davis, 2006; Dawson et al., 2007). SCR captures the electrical conductivity of the skin that is influenced by sweat production and is a marker of sympathetic arousal (Bradley et al., 2001). Typically, fear acquisition leads to increased SCRs for the CS+ (Lonsdorf et al., 2017), that decrease during fear extinction (Sperl et al., 2021). FPS is defined as heightened electromyographic activity of the orbicularis oculi during the startle reflex in response to acoustic stimuli (e.g., white noise burst) when fear is present (i.e., CS+) (Blumenthal et al., 2005). On a neural level, FPS is initiated primarily in brainstem structures (Davis, 2007; Gomez-Nieto et al., 2014; Lee et al., 1996; Lipp, 2006), and modulated by the amygdala and dorsomedial PFC/ACC (Hamm & Weike, 2005). FPS has been related to the aversiveness or negative valence of a stimuli (Bradley et al., 1999). Additional physiological and neuroanatomical outcome measures commonly used include pupil dilation, heart rate, as well as EEG and fMRI measurements (Lonsdorf et al., 2017). Brain imaging has increasingly clarified the mechanisms underlying fear acquisition and extinction. Fear acquisition primarily relies on the "fear network," notably involving the basolateral and centromedial nuclei of the amygdala, the dorsal anterior cingulate cortex (ACC), and the insula (Büchel & Dolan, 2000; Craske et al., 2022; Fullana et al., 2016; Namkung et al., 2022). The brain networks involved in fear extinction learning overlap with these regions and also include the hippocampus and the ventromedial prefrontal cortex (Bouton et al., 2021; Büchel & Dolan, 2000; Fullana et al., 2018; Fullana et al., 2016; Hermann et al., 2016; Kim & Jung, 2006; Milad & Quirk, 2012). In summary, fear conditioning studies offer a valuable experimental framework that deepens our understanding of fear learning and extinction, holding high clinical relevance and translational potential across species and from laboratory research to clinical applications. The different outcome measures, such as physiological responses, imaging data, and self-report questionnaires, highlight various processes and aspects of fear conditioning. Together, they allow for a comprehensive evaluation of fear-related phenomena, facilitating a more nuanced understanding of both the mechanistic underpinnings and the therapeutic implications of anxiety- and stress-related disorders.

Scientific literature and models of PTSD, OCD and anxiety psychopathology highlight that individuals with anxiety- or stress-related disorders exhibit alterations across multiple facets and phases of fear conditioning, including fear acquisition, generalization, extinction, and extinction recall (Cooper et al., 2022; Craske et al., 2014; Duits et al., 2015; Fraunfelter et al., 2022; Lissek et al., 2005; Milad & Quirk, 2012). Findings published up to 2013 have been summarized and

integrated in the influential meta-analysis by Duits et al. (2015) that offers a valuable synthesis of findings and insight in psychopathological mechanisms underlying anxiety. First, it suggests that patients with an anxiety- or stress-related disorder show an exaggerated fear response to safety cues (CS-) rather than an increased response to actual threatening cues (CS+). Second, patients with anxiety- or stress-related disorders exhibit increased fear responses to the conditioned fear stimuli (CS+) during extinction, suggesting impairments in fear extinction learning (Duits et al., 2015). Patients with PTSD were overrepresented in this analysis; however, follow-up analyses suggest that the reported pattern of results holds for both PTSD patients and those with other anxiety-related disorders, including OCD.

The meta-analyses by Duits et al. (2015) has stimulated further research efforts since its release, culminating in many new studies and insights. These studies often support specific changes in fear and safety learning among patients with PTSD, OCD, or other anxiety-related disorders. However, there is significant heterogeneity in the findings both within specific diagnostic categories and across various anxiety- and stress-related disorders. For instance, whereas studies involving patients with PTSD consistently revealed altered fear acquisition compared with healthy participants as indexed with subjective ratings (Diener et al., 2016; Jenewein et al., 2016; Jovanovic et al., 2009) and FPS (Jovanovic et al., 2010; Jovanovic et al., 2009), results are more mixed for other disorders. In the context of OCD, there is mixed evidence regarding alterations in fear acquisition, but stronger evidence exists for impaired extinction learning (Cooper & Dunsmoor, 2021).. Instead, for patients with social anxiety disorder (SAD) a general increased fear/arousal to both the CS+ and CS- has been reported rather than specific increased responding to the CS- (Ahrens et al., 2016). In addition to heterogeneity in results pattern, significant heterogeneity exists in methodology and in the ways, fear learning and extinction have been measured across these studies. Furthermore, studies directly comparing different clinical groups are limited, but they suggest that differential patterns in fear and safety learning may help distinguish between disorders. For instance, one study found that PTSD patients exhibited impaired safety learning and extinction, whereas patients with other anxiety disorders did not show these impairments, as measured through subjective ratings and the FPS response (Acheson et al., 2015). Together, these studies demonstrate substantial heterogeneity in findings across conventional ICD/DSM anxiety- or stress-related disorder categories and underscores the importance of conducting more detailed analyses of fear and safety learning within the spectrum of anxiety and stress-related disorders. It underscores the necessity for an updated meta-analysis to integrate more recent findings and to provide a synthesis of current research in this field. Furthermore, the integration of additional studies enables us to extend the previous analyses. More specifically, this involves integrating results for different

outcomes separately (i.e., expectancy ratings, affect ratings, FPS, and SCR) as opposed to the broader differentiation between physiological and subjective measures in Duits et al. (2015) analyses. Further, we will examine the differences between anxiety disorders, OCD, and PTSD. Additionally, results from extinction recall studies will be integrated and summarized for the first time.

Of note, contemporary models of psychopathology (e.g. HiTOP / RDoC) have gained importance since the initial meta-analysis of Duits et al. (2015). These models highlight the value of looking beyond primary diagnoses, as comorbidity is frequent and within-diagnosis heterogeneity is substantial (Cuthbert & Kozak, 2013; Insel et al., 2010; Kotov et al., 2017). Depressive disorders are the most common comorbid disorders for conditions associated with anxiety- and stress-related disorders (Saha et al., 2020). Consequently, alterations in fear and safety learning that are related to depressive symptoms may also influence the observed patterns of results. Previous research indicated attenuated fear acquisition, as measured with SCR, in individuals with depression (Luo et al., 2024; Otto et al., 2014). Additionally, an increased reactivity to the CS + during extinction learning has been reported in depressed and anhedonic individuals (Wurst et al., 2021; Young et al., 2021). In light of this context, we will include dimensional measures of anxiety and depressive symptom severity, to elucidate the influence of these transdiagnostic dimensions to altered fear and safety learning in the current meta-analysis. Alongside, we will also consider other factors identified in the literature as moderators. We will analyze the influence of US type, CS type, and reinforcement rate as moderators. Previous studies have shown that the US type (e.g., shock, sound) lead to differences in responding. For example, it has been reported that during fear acquisition a scream compared to a shock as the US led to higher reported induced anxiety, lower SCR as well as a more rapid habituation (Ney et al., 2023). Findings reported for different types of CS are rather mixed (e.g., Ahs et al., 2018) with different studies comparing fear relevant (e.g., pictures of spiders or snakes, angry or fearful faces) with neutral stimuli (e.g., geometrical stimuli, neutral faces). A recent systematic review and meta-analysis reported higher SCRs for fear-relevant faces compared to neutral faces (Ney et al., 2022). Lastly, the CS can be paired with the US in 100% of the trials (100% reinforcement rate) or only partially (partial reinforcement). It has been reported that partial reinforcement compared to 100% reinforcement leads to prolonged extinction learning but also reduced response frequency as well as CR amplitudes (for an overview see Lonsdorf et al., 2017). In the lights of these findings, we will include US type, CS type and reinforcement rate as moderators in our analyses. In the present paper we integrate findings of fear and safety learning in stress-related, anxiety-related psychopathology and OCD using a series of meta-analyses to examine differences in fear acquisition, extinction, and extinction recall for the CS+, CS-, and their

discrimination (CS+ minus CS-). Separate meta-analyses will be performed for SCR, FPS, affect and expectancy ratings. Further, the moderating influence of diagnostic category, depressive symptoms, anxiety symptom severity, US type, CS type, and reinforcement rate were considered based on their empirically supported influences.

2 Methods

2.1 Study selection

The review was conducted in adherence to the PRISMA guidelines (Page et al., 2021). Building on the study selection procedure by Duits et al. (2015), we conducted a systematic database search for journal articles published until December 2022 on PubMed. We based our search terms on that of Duits et al. (2015), but modified the search string to include further keywords related to anxiety- and stress-related disorders, ensuring better representation of all anxiety- and stress-related disorders. The final search string was: *"(conditioned OR conditioning) AND (conditioning OR conditioned OR anxiety OR anxious OR fear) AND (PTSD OR post-traumatic stress disorder OR posttraumatic OR phobic OR phobia OR panic OR OCD) AND (patient AND/OR disorder)"*. These terms were required to appear in the title and/or abstract of the papers. No language restrictions were imposed. Figure 1 shows the study flow chart of our selection process. During this process, three independent master students conducted the database search. If inconsistencies between the students occurred, they were checked and resolved by the first author. The search strategy yielded 1660 studies, which were imported into EndNoteData. Each abstract was assessed based on the following inclusion criteria:

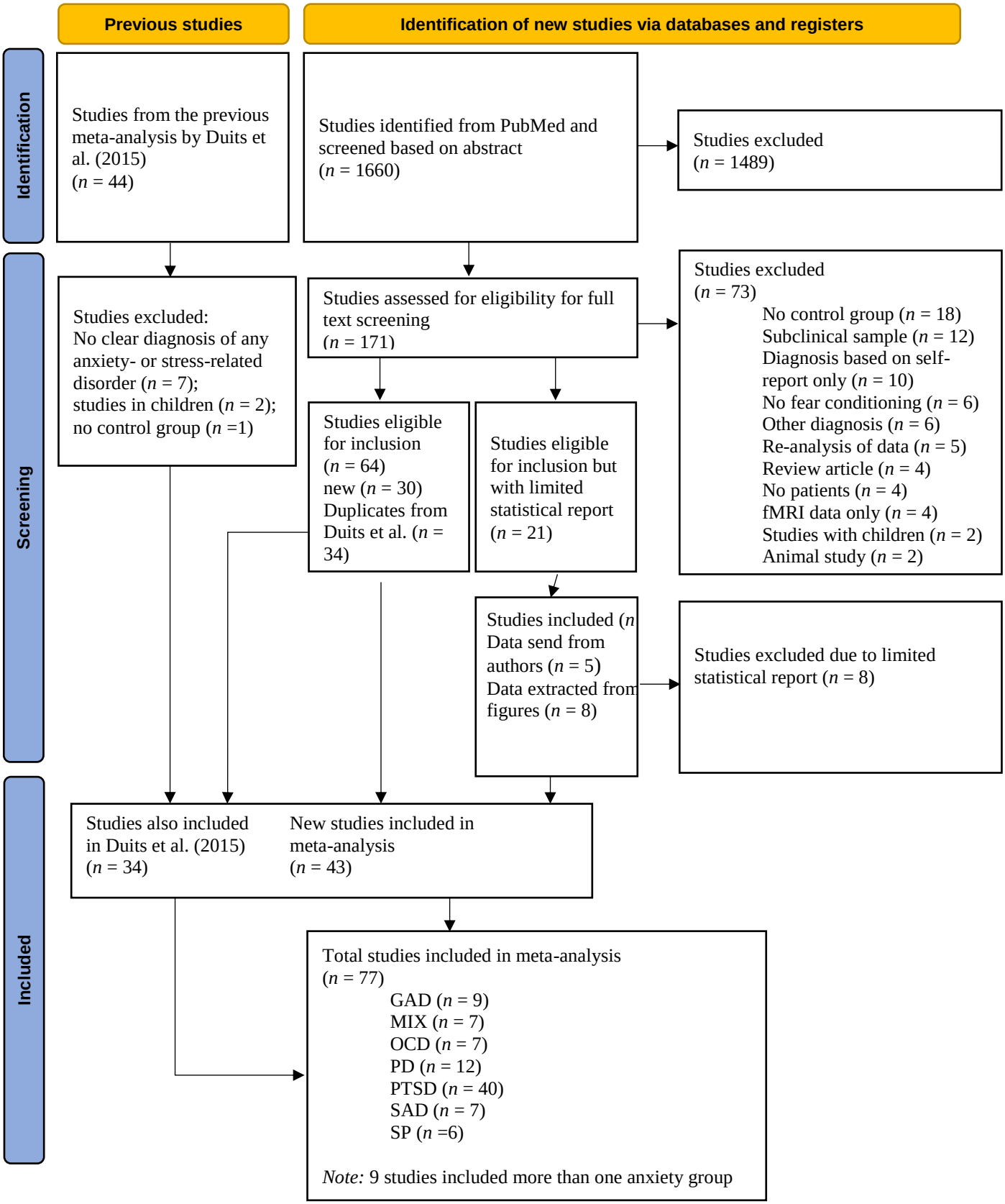
1. Inclusion of adult patients (age ≥ 18 years) with anxiety disorders, OCD, and PTSD, as defined according to DSM criteria (American Psychiatric Association, 2013), as well as healthy participants.
2. Diagnosis of a disorder was established through formal diagnostic interviewing. A self-screening tool was not sufficient.
3. Examination of classical aversive conditioning, at least a phase of fear acquisition had to be present.
4. In addition to fear acquisition, we included studies that investigated fear extinction, and extinction recall, without an experimental manipulation (e.g., reinstatement).
5. Use of subjective (e.g., ratings of US-expectancy or fear) and/or psychophysiological (FPS, SCR, pupil dilation, heart rate) outcome measurements to assess conditioned fear responses.

Following abstract screening, 1489 articles had to be excluded for several reasons such as meta-analysis or review, genetic study, study in animals, pharmacological manipulation, or no fear

conditioning study, leaving 171 studies for full text screening. Of those, 85 studies qualify to be included in the meta-analyses. Because 21 studies did not provide sufficient statistical details to calculate the effect sizes, we reached out to corresponding authors. Although we sent several requests, only five labs provided the requested data. In addition, we were able to extract data from figures in eight studies using WebPlotDigitizer (Rohatgi, 2022). Ultimately, we had to exclude eight studies from the current meta-analysis because the reported statistics did not offer the necessary information to compute or estimate the effect sizes leaving us with 77 included studies in the meta-analyses.

Figure 1

PRISMA Flow diagram for study selection.



Note. OCD = Obsessive compulsive disorder, GAD = Generalized anxiety disorder, MIX = patient sample with various anxiety disorders, PD = Panic disorder, PTSD = Posttraumatic stress disorder, SAD = Social anxiety disorder, SP = Specific phobia.

2.2 Coding of study characteristics and moderators

Relevant study information was extracted (see table 1). Continuous variables (e.g. outcome measurements, reinforcement rate) were directly extracted and not recoded or transformed, other information (e.g. type of CS and US) was coded into categorical variables to study their potentially influence on effect sizes. For meta-regression analyses, similar variables were grouped to form one categorical variable (see table 2).

Table 1

Categorical and Continuous Variables used as Outcomes or Moderators in the meta-analyses

Categorical variables	Type of outcome measurement (SCR, FPS, US expectancy, affect) Primary diagnosis of anxiety- or stress-related disorder (OCD, GAD, MIX, PD, PTSD, SAD, SP) Type of US (airblast, sound, visual, tactile stimulation) Type of CS (Geometric shapes, pictures, lights, faces)
Continuous variables	Reinforcement rate Score of trait anxiety questionnaire (STAI-T) Score of depression questionnaire (BDI)

Note. US = Unconditioned stimulus; CS = Conditioned stimulus; SCR = Skin conductance response; FPS = Fear potentiated startle; STAI = State trait anxiety inventory; BDI = Beck's depression inventory; OCD = Obsessive compulsive disorder, GAD = Generalized anxiety disorder, MIX = patient sample with various anxiety disorders, PD = Panic disorder, PTSD = Posttraumatic stress disorder, SAD = Social anxiety disorder, SP = Specific phobia

Table 2

Grouping of Measures into Categories for Moderators

Variable name	Categories	Included measures
Type of outcome measurement	US expectancy Affect	"US expectancy", "Risk rating" "Valence", "Arousal", "Anxiety", "Fear", "Mood ratings", "Disgust", "Unpleasantness"
Type of US	Sound Visual (Tactile) stimulation	"Scream", "White Noise", "(Aversive) Sound" "(Aversive) pictures", "words" "Shock", "Thermal stimulation", "Painful pressure stimulus", "Aversive odor"
Type of CS	Geometric Shapes Pictures	"Geometric shapes", "colored circles/dot", "circles", "rings" "objects", "words", "pictures", "cartoon characters", "colored lamp", "rooms"

Note. US = Unconditioned stimulus; CS = Conditioned stimulus

2.3 Statistical analyses

2.3.1 Meta-analysis

The data were analyzed using R version 4.3.2 (R Core Team, 2023) and RStudio version 2023.06.1 (RStudio Team, 2023). A *p*-value of .05 was defined as the significance threshold.

To calculate the meta-analytic models, the packages *dmetar* (version 0.1.0; (Harrer et al., 2021), and *meta* (version 6.5-0; (Balduzzi et al., 2019) were used. The code and data used for the analyses are accessible at www.osf/TBA .

As an index of effect sizes, we used the standardized between-group mean difference (SMD). Here, SMD represents the mean difference in fear responses between anxiety patients and healthy participants, standardized by the pooled standard deviation. Thereby, data across studies could be combined and analyzed, independent of their operationalization. We estimated the SMD using Hedges' *g* formula to correct for small sample bias (Hedges, 1981). The way we analyzed our data, positive values indicate stronger responses in anxiety patients, whereas negative values indicate stronger responses in healthy participants. Effect sizes were pooled separately per experimental phase (i.e., acquisition, extinction, and extinction recall), per outcome (i.e., SCR, startle reflex, expectancy ratings, and affect ratings), and per cue (i.e., responses to CS+, responses to CS-, ΔCS). To assess ΔCS (i.e., the difference between CS+ minus CS-) we calculated standardized difference scores per original study, group (i.e., clinical sample and healthy control group), phase, and outcome by calculating hedges *g* between CS+ and CS- and using the *M* of the *SD*s of CS+/CS-. Random effect models were used due to the anticipated substantial between-study heterogeneity and assumed variance in true effects. We expected that potential variation would mainly be introduced in true effects by heterogeneity in the clinical samples and methodological heterogeneity in the experimental protocols. To calculate the heterogeneity variance τ^2, the restricted maximum likelihood estimator was used (Viechtbauer, 2005). The weight given to each study was the inverse of the variance of the effect estimate. To calculate the confidence interval around the pooled effect (using the Q-profile method; (Viechtbauer, 2007), Knapp-Hartung adjustments were used (Knapp & Hartung, 2003). To investigate variation in the included studies, we estimated between-study heterogeneity (i.e., statistical heterogeneity) of the pooled effect sizes and reported Higgins & Thompson's I^2 statistic, which can be interpreted with I^2= 25%: low heterogeneity, I^2= 50%: moderate heterogeneity, and I^2= 75%: substantial heterogeneity (Higgins & Thompson, 2002).

2.3.2 Moderator analyses

To identify possible sources of statistical heterogeneity, we performed follow-up subgroup analyses and meta-regressions if $k \geq 10$ studies reported any of our pre-defined moderator variables for CS+, CS- and ΔCS (Harrer et al., 2021). Subgroup-specific effects were analyzed for categories that had at least 5 studies for each CS+, CS- and ΔCS. The numbers of moderators considered in the analyses differed between phases, outcomes and stimuli. In general, the most moderators could be

analyzed for subjective ratings and acquisition, while the number of studies for physiological outcomes was often insufficient to perform subgroup analyses. Table S3 in the supplement displays the number of studies we could include for the subgroup analyses, and hence which subgroup analyses were conducted. Subgroup analyses included omnibus *Q*-tests indicating whether subgroup differences are large enough not to be explainable by sampling error (Harrer et al., 2021) which were followed-up by calculating effect sizes per level of subgroup separately. Moderators include experimental factors (i.e., US-type, reinforcement rate, and CS-type) and sample characteristics (i.e., diagnostic group, depressive symptoms and anxiety symptoms; see table 1 and 2 for detailed information regarding the moderators). Specifically, to assess potential moderating effects of depressive or anxiety symptoms, the SMD between patient group and healthy control group were calculated within each primary study, if data of trait anxiety, state anxiety (STAI-T and STAI-S scores; Spielberger et al., 1983), or depression questionnaire, i.e. BDI (Beck et al., 1996), HAM-D (Hamilton, 1960), CES-D (Radloff, 1977), ADS (Meyer & Hautzinger, 2001), IDS-SR (Rush et al., 1986), DASS-D (Nilges & Essau, 2021) were reported. This SMD of depressive, state or trait anxiety symptoms within the primary study represents the extent to which the patient group and healthy control group differ in the respective symptom dimension and was used to examine potential influences on task effects (i.e., fear conditioning processes). This enables testing potential moderating effects of dimensional clinical measures across different questionnaires used to assess the same construct using meta-regression. Our main interest was to compare anxiety- and stress-related disorder categories. Due to the overrepresentation of PTSD studies in contrast to all other anxiety-related disorders and OCD, we performed subgroup analyses and compared healthy participants vs. PTSD studies as well as healthy participants vs. all other anxiety-related disorders including OCD, following the approach of Duits et al. If number of studies within each single diagnostic category was at least 5, we also performed diagnose specific subgroup analyses. If more than one experimental condition was reported in a primary study, i.e., if different anxiety groups were included (e.g. see Grillon et al., 2009; Klahn et al., 2017) the comparisons were included twice but the healthy participant sample was divided by 2 to correct for multiple testing within one group following the methodological approach of the meta-analysis by Duits et al. (Duits et al., 2015).

2.3.3 Outlier analysis and small sample bias assessment

To assess the robustness of our results, we performed a basic outlier analysis using the function *find.outliers* which is included in the *dmetar* package (Harrer et al., 2021). Here, an outlier is detected when its confidence interval does not overlap with the confidence interval of the pooled effect. In our results section, we report the results after outlier analysis and after exclusion of

outlying studies. Since the number of outliers was relatively high and to allow the reader to assess the model without outlier removal, we presented the results of the main analyses without outlier removal in the supplement S2, highlighting discrepancies with another font color and refer in the text to discrepancies between these two types of analysis. Additionally, especially in the area of moderator analyses, many models could not be computed due to the insufficient number of studies available after outlier exclusion during certain phases or for specific stimuli. On one hand, the relatively conservative standards we set for the inclusion of studies ensure the robustness of the results and reduce the overall already high number of findings to those with higher robustness. On the other hand, by excluding outliers in the moderator analyses, we also reduce variance that could potentially be explained by our moderators. Thus, all moderator analyses for the full model with all datasets are included in the supplement, and we provide a brief description of the model in the paper. In addition, we examined the presence of publication bias focusing on small-study effects by assessing asymmetry in funnel plots and applying Egger's regression test (Egger et al., 1997) of the intercept, to quantify the presence of funnel plot asymmetry. The intercept of the linear model represents the relationship of the standardized effect sizes to precision; in the absence of publication bias, the intercept is expected to be zero (Harrer et al., 2021). The presence of potential bias was indicated in the respective result sections. Moreover, we conducted *p*-curve analyses, since it is assumed that they represent more accurately the true effect size behind the data (Simonsohn et al., 2014). We used the function that is included in the *dmetar* package (Harrer et al., 2021). Due to its sensitivity to between-study heterogeneity, it's proposed to only use *p*-curve when heterogeneity is small to moderate (as a rule of thumb, i.e. $I^2 < 50\%$) (van Aert et al., 2016).

3 Results

In total, we included 77 studies, published between 1986 and 2022, resulting in data of 2052 patients with anxiety disorders and 3258 healthy participants (mean age = 34.45, 49.71 % female). The majority of studies ($n = 40$) included patients with PTSD, and $n = 12$ studies compared patients with PD with healthy participants (HC). For all other anxiety disorders less than ten studies were included in the meta-analysis (n = 9 GAD, $n = 7$ OCD, $n = 7$ SAD, $n = 6$ SP). In addition, $n = 7$ studies included multiple anxiety disorders without specifying the exact sample size per diagnostic category (Mixed anxiety group, MIX). Note that nine studies included more than one diagnostic group, why the number of diagnostic groups is higher than the number of included studies (i.e. Acheson et al., 2015; Fyer et al., 2020; Grillon et al., 2009; Klahn et al., 2017; Marin et al., 2020; Otto et al., 2014; Pohlchen et al., 2020; Rabinak et al., 2017; Tinoco-Gonzalez et al., 2015). For an overview of the included studies and their key experimental characteristics, see table 3. For a more

detailed overview of the included studies (i.e., details regarding age, gender distribution, anxiety, and depressive symptoms), see table S1 in the supplement.

Not all studies included in our analysis examined all experimental phases or used the same behavioral and physiological measures, leading to variations in the number of observations in each analysis (986 to 2500 data points, i.e., sample size in primary study). We analyzed data from 18 to 54 studies for the acquisition phase (see table 4), 5 to 27 studies for the extinction phase (see table 5), and 2 to 15 studies for the extinction recall phase (see table 6).

Table 3

Overview of Included Studies

Study	Diagnostic Group	Total Sample size *N* (Subgroups included in the meta-analysis)	Type of CS	Type of US	Type of Outcome Measurement	Experimental paradigm
Acheson et al., 2015	PTSD, Mixed Anxiety	1002 (n=42 PTSD, n=37 MIX, n=923 HC)	Geometrical Shapes	Airblast	FPS; Rating (US Expectancy)	1-Day Paradigm Acquisition; Extinction
Ahrens et al., 2016	SAD	55 (n=29 SAD, n=26 HC)	Faces	Visual (Fearful Face) and Sound (Scream)	HR; Rating (CS Arousal; CS Valence; US Expectancy); SCR	1-Day Paradigm Habituation; Acquisition; Generalization
Apergis-Schoute et al., 2017	OCD	78 (n=43 OCD, n=35 HC)	Faces	Stimulation (Shock)	fMRI; SCR	1-Day Paradigm Acquisition; Reversal
Armstrong et al., 2021	PTSD	51 (n=20 PTSD, n=31 TE)	Geometrical Shapes	Sound (Scream)	Rating (CS Anxiety; US Expectancy)	1-Day Paradigm Habituation; Acquisition; Re-Acquisition; Extinction
Berg et al., 2021	PTSD	39 (n=20 PTSD, n=19 TE)	Geometrical Shapes	Stimulation (Shock)	fMRI; Rating (US Expectancy)	1-Day Paradigm Habituation; Acquisition; Generalization
Blechert et al. 2007	PTSD	91 (n=26 PTSD, n=34 HC, n=21 TE)	Pictures	Stimulation (Shock)	Rating (CS Valence; US Expectancy); SCR	1-Day Paradigm Habituation; Acquisition; Extinction
Britton et al., 2013	Mixed Anxiety	49 (n=18 MIX, n=31 HC)	Faces	Visual (Fearful Face) and Sound (Scream)	fMRI; Rating (US Expectancy)	2-Day Paradigm Day 1: Pre-Acquisition; Acquisition; Extinction Day 2: Extinction Recall; Generalization
Cano et al., 2021	OCD	36 (n=18 OCD, n=18 HC)	Pictures	Stimulation (Shock)	fMRI; SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall

Diener et al., 2016	PTSD	41 (n=14 PTSD, n=13 HC, n=14 TE)	Geometrical Shapes	Visual (Aversive Pictures)	fMRI; Rating (CS Arousal; CS Valence; US Expectancy)	1-Day Paradigm Habituation; Acquisition; Extinction
Elsner et al., 2022	OCD	73 (n=41 OCD, n=32 HC)	Faces	Stimulation (Shock)	SCR	1-Day Paradigm Acquisition; Reversal
Fyer et al., 2020	OCD, SAD	146 (n=41 OCD, n=41 SAD, n=64 HC)	Lights	Stimulation (Shock)	SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall; Fear Renewal
Garfinkel et al., 2014	PTSD	28 (n=14 PTSD, n=14 TE)	Lights	Stimulation (Shock)	fMRI; SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall; Fear Renewal
Greenberg et al., 2013	GAD	57 (n=32 GAD, n=25 HC)	Geometrical Shapes	Stimulation (Shock)	fMRI; Pupil Data; Rating (US Expectancy)	1-Day Paradigm Generalization
Grillon & Morgan, 1999	PTSD	24 (n=12 PTSD, n=12 TE)	Lights	Stimulation (Shock)	FPS	2-Day Paradigm Day 1: Pre-Acquisition; Acquisition; Extinction Day 2: Pre-Acquisition; Acquisition; Extinction
Grillon et al., 1998	PTSD	65 (n=34 PTSD, n=14 HC, n=17 TE)	Lights	Stimulation (Shock)	FPS	1-Day Paradigm Acquisition
Grillon et al., 2008	PD	48 (n=24 PD, n=24 HC)	Geometrical Shapes	Sound (White Noise, Smoke Alarm, Female Scream + Picture)	FPS; Rating (US Expectancy)	1-Day Paradigm Acquisition
Grillon et al., 2009	GAD, PTSD	68 (n=18 GAD, n=16 PTSD, n=34 HC)	Geometrical Shapes	Sound (White Noise, Smoke Alarm, Female Scream + Picture)	FPS	1-Day Paradigm Habituation; Acquisition

Helpman et al., 2016	PTSD	58 (n=30 PTSD, n=28 TE)	Lights	Stimulation (Shock)	fMRI; SCR	2-Day Paradigm Day 1: Acquisition; Extinction Day 2: Extinction Recall
Hermann et al., 2002	SAD	33 (n=14 SAD, n=19 HC)	Faces	Stimulation (Aversive Odor)	EMG; FPS; HR; Rating (CS Arousal; CS Valence; US Expectancy); SCR	1-Day Paradigm Habituation; Acquisition; Extinction
Jenewein et al., 2016	PTSD	40 (n=19 PTSD, n=21 HC)	Geometrical Shapes	Stimulation (Thermal Stimuli)	Rating (CS Fear; US Expectancy); SCR	1-Day Paradigm Acquisition
Jovanovic et al., 2009	PTSD	55 (n=13 low-PTSD, n=14 high-PTSD n=28 HC)	Lights	Airblast	FPS; Rating (US Expectancy)	1-Day Paradigm Habituation; Acquisition
Jovanovic et al., 2010a	PTSD	67 (n=14 PTSD, n=53 TE)	Geometrical Shapes	Airblast	FPS	1-Day Paradigm Habituation; Acquisition; Fear Inhibition Transfer Test
Jovanovic et al., 2010b	PTSD	90 (n=29 PTSD, n=61 TE)	Geometrical Shapes	Airblast	FPS; Rating (US Expectancy)	1-Day Paradigm Habituation; Acquisition
Jovanovic et al., 2011	PTSD	54 (n=16 PTSD, n=38 TE)	Geometrical Shapes	Airblast	FPS; Rating (US Expectancy)	1-Day Paradigm Habituation; Acquisition
Jovanovic et al., 2013	PTSD	41 (n=20 PTSD, n=21 TE)	Geometrical Shapes	Airblast	FPS	1-Day Paradigm Acquisition; Extinction
Kaczkurkin et al., 2017	PTSD	40 (n=20 PTSD, n=20 TE)	Geometrical Shapes	Stimulation (Shock)	fMRI; Rating (US Expectancy)	1-Day Paradigm Pre-Acquisition; Acquisition; Generalization
Kircher et al., 2013	PD	84 (n=42 PD, n=42 HC)	Geometrical Shapes	Sound (White Noise)	fMRI; Rating (CS Arousal; CS Valence)	1-Day Paradigm Habituation; Acquisition; Extinction
Klahn et al., 2017	PD, SP	60 (n=20 PD, n=20 SP, n=20 HC)	Faces	Visual (Video of a Monster) and Sound (Scream)	MEG; Rating (CS Arousal; CS Valence; US Expectancy)	1-Day Paradigm Acquisition

Lange et al., 2019	SP	94 (n=46 SP, n=48 HC)	Geometrical Shapes	Stimulation (Shock)	fMRI; Rating (US Expectancy; CS Fear)	3-Day Paradigm Day 1: Pre-Acquisition; Acquisition; Generalization Day 2: Extinction Day 3: Extinction Recall
Linnman et al., 2011	PTSD	43 (n=19 PTSD, n=24 TE)	Lights	Stimulation (Shock)	fMRI; Rating; SCR	1-Day Paradigm Acquisition
Lissek et al., 2008	SAD	38 (n=20 SAD, n=18 HC)	Faces	Visual (Critical Facial Expression) and Sound (Insults)	FPS; Rating (CS Anxiety; CS Valence; CS Arousal)	1-Day Paradigm Pre-Acquisition; Acquisition; Extinction
Lissek et al., 2009	PD	48 (n=24 PD, n=24 HC)	Pictures	Stimulation (Shock)	FPS; Rating (CS Anxiety; US Expectancy)	2-Day Paradigm Day 1: Habituation; Acquisition Day 2: Retention/Extinction
Lissek et al., 2010	PD	38 (n=19 PD, n=19 HC)	Geometrical Shapes	Stimulation (Shock)	FPS; Rating (CS Anxiety; US Expectancy)	1-Day Paradigm Habituation; Acquisition; Generalization
Lissek et al., 2014	GAD	48 (n=22 GAD, n=26 HC)	Geometrical Shapes	Stimulation (Shock)	FPS; Rating (CS CS Anxiety; US Expectancy)	1-Day Paradigm Pre-Acquisition; Acquisition; Generalization
Lueken et al., 2014	PD	120 (n=60 PD, n=60 HC)	Geometrical Shapes	Sound (Aversive)	fMRI; Rating (CS Arousal; CS Valence)	1-Day Paradigm Habituation; Acquisition; Extinction
Marin et al., 2016	PTSD	65 (n=24 PTSD, n=21 HC, n=20 TE)	Lights	Stimulation (Shock)	fMRI; SCR	2-Day Paradigm Day 1: Acquisition; Extinction Day 2: Extinction Recall
Marin et al., 2017	Mixed Anxiety	82 (n=61 MIX, n=21 HC)	Lights	Stimulation (Shock)	fMRI; SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall

Marin et al., 2020	GAD, PD, SP, SAD	114 (n=27 GAD, n=18 PD, n=28 SAD, n=20 SP, n=21 HC)	Lights	Stimulation (Shock)	fMRI; SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall
McLaughlin et al., 2015	OCD	46 (n=31 OCD, n=15 HC)	Lights	Stimulation (Shock)	SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall; Fear Renewal
Michael et al., 2007	PD	72 (n=33 PD, n=39 HC)	Pictures	Stimulation (Shock)	Rating (CS Valence); SCR	1-Day Paradigm Habituation; Acquisition
Michopoulos et al., 2017	PTSD	63 (n=24 PTSD, n=29 TE)	Geometrical Shapes	Airblast	FPS	1-Day Paradigm Acquisition; Extinction
Milad et al., 2008	PTSD	28 (n=7 PTSD, n=14 HC, n=7 TE)	Lights	Stimulation (Shock)	SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall
Milad et al., 2009	PTSD	31 (n=16 PTSD, n=15 HC)	Lights	Stimulation (Shock)	fMRI; FPS	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall
Milad et al., 2013	OCD	42 (n=21 OCD, n=21 HC)	Pictures	Stimulation (Shock)	fMRI; SCR	2-Day Paradigm Day 1: Acquisition; Extinction Day 2: Extinction Recall
Miller et al., 2011	PTSD	63 (n=32 PTSD, n=31 HC)	Geometrical Shapes	Stimulation (Shock)	FPS	2-Day Paradigm Day 1: Habituation, Acquisition Day 2: Habituation, Acquisition
Morey et al., 2015	PTSD	67 (n=32 PTSD, n=35 HC)	Faces	Stimulation (Shock)	fMRI; Rating (US Expectancy)	1-Day Paradigm Habituation; Pre-Acquisition; Acquisition; Generalization
Morey et al., 2020	PTSD	36 (n=17 PTSD, n=19 HC)	Pictures	Stimulation (Shock)	fMRI; Rating (US Expectancy); SCR	2-Day Paradigm Day 1: Acquisition Day 2: Recognition Memory Test

Morgan, III., et al., 1995	PTSD	19 (n=9 PTSD, n=10 HC)	Lights	Stimulation (Shock)	FPS	1-Day Paradigm Habituation; Acquisition
Nanbu et al., 2010	OCD	60 (n=39 OCD, n=21 HC)	Lights	Stimulation (Shock)	EEG; SCR	1-Day Paradigm Habituation; Acquisition; Extinction
Neueder et al., 2019	PD	43 (n=21 PD, n=22 HC)	Pictures	Stimulation (Shock)	FPS; Rating (CS Anxiety; CS Arousal; CS Valence; US Expectancy); SCR	1-Day Paradigm Acquisition; Generalization
Norrholm et al., 2011	PTSD	127 (n=49 PTSD, n=78 TE)	Geometrical Shapes	Aairblast	FPS; Rating (US Expectancy)	1-Day Paradigm Acquisition; Extinction
Norrholm et al., 2013	PTSD	270 (n=98 PTSD, n=172 HC)	Geometrical Shapes	Airblast	FPS; Rating (US Expectancy)	1-Day Paradigm Acquisition; Extinction
Orr et al., 2000	PTSD	33 (n=15 PTSD, n=18 HC)	Geometrical Shapes	Stimulation (Shock)	FPS; HR; SCR	1-Day Paradigm Habituation; Acquisition; Extinction
Otto et al., 2014	PD, PTSD	146 (n=21 PD, n=16 PTSD, n=96 HC)	Geometrical Shapes	Stimulation (Shock)	Rating (US Expectancy); SCR	2-Day Paradigm Day 1: Habituation; Acquisition Day 2: Extinction
Peri et al., 2000	PTSD	86 (n=36 PTSD, n=30 HC, N=20 TE)	Lights	Sound (White Noise)	HR; SCR	1-Day Paradigm Habituation; Acquisition; Extinction
Pineles et al., 2016	PTSD	32 (n=16 PTSD, n=16 TE)	Geometrical Shapes	Stimulation (Shock)	SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall
Pitman et al., 1986	GAD	40 (n=20 GAD, n=20 HC)	Faces	Stimulation (Shock)	SCR	1-Day Paradigm Habituation; Acquisition; Extinction (immediate)
Pittig et al., 2020	Mixed Anxiety	80 (n=40 MIX, n=40 HC)	Geometrical Shapes	Stimulation (Shock)	Rating (US Expectancy); SCR	1-Day Paradigm Habituation; Acquisition

Pöhlchen et al., 2020	Mixed Anxiety, PTSD	129 (n=21 MIX, n=73 PTSD, n=35 HC)	Geometrical Shapes	Stimulation (Shock), Airblast	FPS; Pupil dilation; Rating (US Expectancy); SCR	2-Day Paradigm 1 Day: Acquisition; Extinction Day 2: Extinction Recall
Rabinak et al., 2017	PTSD, SAD	44 (n=13 PTSD, n=16 SAD, n=15 HC)	Lights	Sound (White Noise)	Rating (Shock Expectancy)	2-Day Paradigm Day 1: Acquisition; Extinction Day 2: Extinction Recall
Schwarzmeier et al., 2019	PD	20 (n=10 PD, n=10 HC)	Faces	Sound (Scream)	fMRI; Rating (CS Arousal; CS Valence); SCR	3-Day Paradigm Day 1: Habituation; Acquisition Day 2: Extinction Day 3: Extinction Recall
Schweckendiek et al., 2011	SP	29 (n=15 SP, n=14 HC)	Geometrical Shapes	Visual (pictures)	fMRI; Rating (CS Arousal; CS Disgust; CS Fear; CS Valence; US Expectancy); SCR	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction
Seo et al., 2022	PTSD	126 (n=63 PTSD, n=63 TE)	Lights	Stimulation (Shock)	fMRI; SCR; US Expectancy	1-Day Paradigm Habituation; Acquisition; Extinction
Shechner et al., 2015	Mixed Anxiety	52 (n=22 MIX, n=30 HC)	Pictures	Sound (Bell, aversive)	FPS; Rating (CS Fear); SCR	1-Day Paradigm Pre-Acquisition; Acquisition; Extinction
Steiger et al., 2015	PTSD	37 (n=12 PTSD, n=11 HC, n=14 TE)	Geometrical Shapes	Stimulation (Shock)	fMRI; Rating (CS Arousal; CS Valence; US Expectancy)	2-Day Paradigm Day 1: Habituation; Acquisition; Extinction Day 2: Extinction Recall
Thome et al., 2018	PTSD	90 (n=30 PTSD, n=30 HC, n=30 TE)	Geometrical Shapes	Stimulation (Shock)	FPS; Rating (US Expectancy)	1-Day Paradigm Pre-Acquisition; Acquisition; Generalization
Tinoco-González et al., 2015a	GAD	58 (n=28 GAD, n=30 HC)	Geometrical Shapes	Stimulation (Shock)	FPS; Rating (CS Anxiety; CS Arousal; US Expectancy)	1-Day Paradigm Habituation; Acquisition; Generalization
Tinoco-González et al., 2015b	PD, SAD	48 (n=16 PD, n=16 SAD, n=16 HC)	Faces	Visual (Critical Facial	FPS; HR; Rating (CS Anxiety; CS Arousal; CS Valence)	1-Day Paradigm Habituation; Acquisition; Reminder; Extinction

				Expression) and Sound (Insults)		
Veit et al., 2002	SP	11 (n=4 SP, n=7 HC)	Faces	Stimulation (Painful Pressure)	fMRI; Rating (US Expectancy); SCR	1-Day Paradigm Habituation; Acquisition; Extinction
Via et al., 2018	GAD	83 (n=27 GAD, n=56 HC)	Geometrical Shapes	Sound (Noise Burst)	fMRI; Rating (CS Arousal; CS Valence)	1-Day Paradigm Habituation; Acquisition
Vriends et al., 2012	SP	72 (n=33 SP, n=39 HC)	Pictures	Visual (pictures and words)	Rating (CS Fear; CS Valence; US Expectancy)	1-Day Paradigm Acquisition
Wang et al., 2021	GAD	58 (n=29 GAD, n=29 HC)	Pictures	Visual (pictures, unpleasant)	Rating (US Expectancy)	1-Day Paradigm Acquisition; Generalization
Wessa et al., 2007	PTSD	44 (n=14 PTSD, n=15 HC, n=15 TE)	Geometrical Shapes	Visual (pictures, trauma-specific)	EEG; EOG; FPS; Rating (CS Arousal; CS Valence; US Expectancy); SCR	1-Day Paradigm Habituation; Acquisition; Extinction
Wicking et al., 2016	PTSD	54 (n=18 PTSD, n=18 HC, n=18 TE)	Geometrical Shapes	Stimulation (Shock)	fMRI; Rating (CS Arousal; CS Valence; US Expectancy); SCR	3-Day Paradigm Day 1: Habituation; Acquisition Day 2: Extinction Day 3: Extinction Recall
Wurtz et al., 2016	PTSD	35 (n=18 PTSD, n=17 HC)	Pictures	Stimulation (Shock)	Rating (CS Valence; US Expectancy); SCR	1-Day Paradigm Habituation; Acquisition; Extinction

Note. CS = Conditioned Stimulus, ECG = Electrocardiography, EEG = Electroencephalography, EMG = Electromyography, EOG = Electroocoulography, GAD = Generalized Anxiety Disorder, fMRI = functional Magnetic Resonance Imaging, FPS = Fear-potentiated startle, HC = Healthy Controls, HR = Heart Rate, MEG = Magnetoencephalography, MIX = Mixed Anxiety Disorders, OCD = Obsessive-Compulsive Disorder, PD = Panic Disorder, PTSD = Posttraumatic Stress Disorder, SAD = Social Anxiety Disorder, SCR = Skin Conductance Response; SP = Specific Phobia, TE = Trauma Exposed, US = Unconditioned Stimulus. Outcome measurements are grey colored when they were reported in the primary study but not investigated by the present meta-analysis.

3.1 Acquisition phase

Table 4 displays the integrated effect size, 95% confidence interval, the estimation of between-study heterogeneity I^2 and statistical values for the observed group differences for CS+, CS-, as well as ΔCS separately for the different outcome measurements during the acquisition phase, after outlier analyses. Forest plots visualizing the effect sizes of the contributing primary studies and the heterogeneity are presented separately by outcome for CS+ and CS-, as well as for ΔCS in supplementary material accompanying this article in figure S1.

Results suggest no difference in response to the CS+, on physiological level (SCR and FPS), or regarding US expectancy (all $p \geq .20$). However, anxiety patients seem to show increased negative affect towards the CS+ compared with healthy participants as suggested by affect ratings ($g = 0.52$; 95% CI: 0.30-0.74; $p < .001$). Regarding the CS-, results showed significant higher responses in anxiety patients compared to healthy participants across multiple outcome measurements, i.e., FPS ($g = 0.29$; 95% CI: 0.16-0.42; $p < .001$), US expectancy ($g = 0.29$; 95% CI: 0.14-0.43; $p < .001$), and affect ratings ($g = 0.36$; 95% CI: 0.20-0.52; $p < .001$). Patients and healthy participants did not differ regarding their SCR in response to the CS-. Regarding ΔCS, results suggest no differences between patients and healthy participants in any of our outcome measurements (all $p \geq .066$).

In sum, it seems that anxiety patients compared to healthy participants do not respond significantly differently to a threat associated stimulus on a psychophysiological and cognitive level, but they subjectively indicate to experience increased negative affect to a threatening stimulus, i.e. CS+. In contrast, multiple outcome measures suggest that anxiety patients seem to associate an actual safe stimulus (CS-) with increased fear on cognitive, subjective and partly psychophysiological levels (FPS). This implies that anxiety patients may have impairments in regulating their fear in response to a safe stimulus. The results for ΔCS did not indicate any consistent differences in fear acquisition across any of the outcome measures. Heterogeneity was low to moderate for the CS- analyses (I^2 = 0- 46%), moderate to high for the CS+ (I^2 = 28-61 %), and with one exception mostly high for the ΔCS (I^2 = 0-93 %), underscoring the need for additional moderator analyses.

The results of the full model, which includes all studies without excluding outliers (see table S2 in the supplement), are consistent, except for an additional effect observed for ΔCS measured with FPS, indicating a reduced differentiation between CS+ and CS- in patients with stress-related and anxiety-related disorders compared to healthy participants. However, as expected, there was substantial heterogeneity across all stimuli in all outcome measurements (I^2 = 8-93%).

Table 4

Statistical model parameters of the acquisition phase

Outcome measure	Stimulus	Model	Number of studies and participants	*g*	95%CI	*p*	I^2
Skin Conductance Response	CS+	Main model after outlier removal	28 n_{pat} = 678, n_{HC} = 751	-0.02	-0.19-0.15	.853	0.52
		Subgroup analysis after outlier removal				.265	
		▪ Anxiety disorders and OCD vs. HC	15 n_{pat} = 444, n_{HC} = 449	-0.09	-0.34-0.16	.450	0.61
		▪ PTSD vs. HC	13 n_{pat} =234, n_{HC} = 302	0.09	-0.16-0.35	.439	0.33
	CS-	Main model after outlier removal	25 n_{pat} = 615, n_{HC} = 685	-0.01	-0.16-0.13	.830	0.24
		Subgroup analysis after outlier removal				**.**110	
		▪ Anxiety disorders and OCD vs. HC	13 n_{pat} = 400, n_{HC} = 407	-0.10	-0.30-0.10	.308	0.34
		▪ PTSD vs. HC	12 n_{pat} = 215, n_{HC} = 278	0.11	-0.09-0.31	.255	0.0
	ΔCS	Main model after outlier removal	29 n_{pat} = 754, n_{HC} = 776	0.50	-0.04-1.04	.066	0.93
		Subgroup analysis after outlier removal				**.035***	
		▪ Anxiety disorders and OCD vs. HC	16 n_{pat} = 539, n_{HC} = 498	0.92	0.06-1.78	**.038***	0.95
		▪ PTSD vs. HC	13 n_{pat} = 215, n_{HC} = 278	-0.04	-0.47-0.40	.862	0.75
Fear Potentiated Startle	CS+	Main model after outlier removal	20 n_{pat} = 468, n_{HC} = 1533	0.01	-0.17-0.19	.904	0.44
		Subgroup analysis after outlier removal				.820	
		▪ Anxiety disorders and OCD vs. HC	9 n_{pat} = 245, n_{HC} = 251	-0.01	-0.27-0.25	.901	0.30
		▪ PTSD vs. HC	11 n_{pat} = 223, n_{HC} = 1282	0.02	-0.27-0.32	.853	0.55
	CS-	Main model after outlier removal	18 n_{pat} = 410, n_{HC} = 576	0.29	0.16-0.42	**<.001*****	0.00
		Subgroup analysis after outlier removal				.871	

		▪ Anxiety disorders and OCD vs. HC	9 n_{pat} = 245, n_{HC} = 251	0.30	0.09-0.51	**.011***	0.0
		▪ PTSD vs. HC	9 n_{pat} = 165, n_{HC} = 325	0.28	0.06-0.50	**.018***	0.0
	ΔCS	Main model after outlier removal	20 n_{pat} = 433, n_{HC} = 624	-0.12	0.25-0.01	.077	0.0
		Subgroup analysis after outlier removal				.834	
		▪ Anxiety disorders and OCD vs. HC	8 n_{pat} = 227, n_{HC} = 217	-0.13	-0.32-0.05	.129	0.0
		▪ PTSD vs. HC	12 n_{pat} = 206, n_{HC} = 407	-0.11	-0.32-0.11	.292	0.17
US Expectancy Rating	CS+	Main model after outlier removal	29 n_{pat} = 615, n_{HC} = 799	0.005	-0.13-0.14	.948	0.28
		Subgroup analysis after outlier removal				.535	
		▪ Anxiety disorders and OCD vs. HC	8 n_{pat} = 221, n_{HC} = 202	-0.06	-0.46-0.34	.738	0.61
		▪ PTSD vs. HC	21 n_{pat} = 394, n_{HC} = 597	0.05	-0.08-0.19	.423	0.0
	CS-	Main model after outlier removal	26 n_{pat} = 595, n_{HC} = 755	0.29	0.14-0.43	**<.001****	0.30
		Subgroup analysis after outlier removal				**.002****	
		▪ Anxiety disorders and OCD vs. HC	9 n_{pat} = 250, n_{HC} = 228	0.58	0.26-0.90	**.003****	0.55
		▪ PTSD vs. HC	17 n_{pat} = 345, n_{HC} = 527	0.12	0.02-0.22	**.021***	0.0
	ΔCS	Main model after outlier removal	29 n_{pat} = 644, n_{HC} = 825	-0.25	-0.73-0.24	.308	0.85
		Subgroup analysis after outlier removal				.102	
		▪ Anxiety disorders and OCD vs. HC	9 n_{pat} = 250, n_{HC} = 228	-1.16	-2.90-0.58	.162	0.90
		▪ PTSD vs. HC	20 n_{pat} = 394, n_{HC} = 597	0.09	-0.20-0.38	.519	0.75
Affect Rating	CS+	Main model after outlier removal	33 n_{pat} = 468, n_{HC} = 611	0.52	0.30-0.74	**<.001****	0.61
		Subgroup analysis after outlier removal				**.006****	

		▪ Anxiety disorders and OCD vs. HC	19 n_{pat} = 341, n_{HC} = 383	0.73	0.41-1.04	**<.001*** **	0.69
		▪ PTSD vs. HC	14 n_{pat} = 127, n_{HC} = 228	0.22	-0.01-0.46	.061	0.0
	CS-	Main model after outlier removal	34 n_{pat} = 550, n_{HC} = 741	0.36	0.20-0.52	**<.001*** **	0.46
		Subgroup analysis after outlier removal				.066	
		▪ Anxiety disorders and OCD vs. HC	19 n_{pat} = 405, n_{HC} = 496	0.47	0.26-0.67	<**.001*** **	0.51
		▪ PTSD vs. HC	15 n_{pat} = 145, n_{HC} = 245	0.17	-0.11-0.45	.212	0.32
	ΔCS	Main model after outlier removal	54 n_{pat} = 960, n_{HC} = 1190	0.17	-0.17-0.52	.319	0.86
		Subgroup analysis after outlier removal				.744	
		▪ Anxiety disorders and OCD vs. HC	36 n_{pat} = 763, n_{HC} = 856	0.22	-0.31-0.75	.403	0.90
		▪ PTSD vs. HC	18 n_{pat} = 197, n_{HC} = 334	0.13	-0.09-0.35	.226	0.23

Note: n_{pat} = sample size patients, n_{HC} = sample size healthy participants, g = Hedge's g, CI = confidence interval, PTSD = Posttraumatic stress disorder. Overall *p* values for subgroup analyses are based on *Q* tests testing the H_0 that effect sizes of all subgroups are equal. Text written in grey: subgroup specific effects when subgroup analyses were non-significant. *p < .05, **p < .01, ***p < .001

3.1.1 Moderator Analyses: Subgroup analyses PTSD vs. anxiety disorders and OCD

Results for the subgroup analyses based on *diagnostic group* (PTSD vs. anxiety disorders and OCD) are also presented in table 4. Subgroup-specific effects were only investigated if the results for the subgroup analyses were significant. Regarding the CS+, there was a significant difference between groups for affect ratings (p = .006). We obtained a significant subgroup-specific effect for the anxiety disorders and OCD group (g = 0.73, p < .001), suggesting significant higher affect ratings for the CS+ for anxiety and OCD patients compared with healthy participants and the PTSD group which did not differ from healthy participants. This suggests that individuals with an anxiety disorder or OCD may give increased affect ratings to the CS+, an effect that is not observed in PTSD. Regarding the CS-, results showed a significant difference between subgroups in US expectancy ratings (p = .002). Follow-up analyses revealed subgroup-specific effects for both groups (PTSD: g = 0.12, p = .021; Anxiety disorders and OCD: g = 0.58, p = .003), suggesting significantly higher US expectancy rating towards the CS- in both groups. Notably, patients with anxiety disorders and OCD demonstrated significantly stronger effects, reflecting larger effect sizes

and higher US expectancy ratings to the CS- compared to those with PTSD. Regarding the ΔCS, we found a significant difference between groups in SCR responses (p = .035). For PTSD, the subgroup-specific effect was not significant, indicating no difference in ΔCS responding compared to healthy individuals. There was a significant effect for the group anxiety disorders and OCD (g = 0.92, p = .036), suggesting a significant greater difference of SCR responding in ΔCS for anxiety and OCD patients compared to healthy participants.

In summary, our findings indicate that for most measures the pattern of results observed during the acquisition phase is not significantly different for patients with OCD and anxiety compared to those with PTSD. However, some exceptions have to be noted where the two groups differed. Follow-up analyses support significant differences between anxiety and OCD for ΔCS for the SCR responding, for CS- for expectancy ratings, and for CS+ and CS- for affect ratings. Notably, in all these cases, effect sizes and differences were larger for the anxiety/OCD group compared to healthy controls, while they were smaller or even non-significant for the PTSD group. Overall, there was a highly varying between-study heterogeneity in both groups across almost all outcome measures (I^2 = 0. 95%) indicating large variability among studies, underscoring the importance of investigating potential moderators.

Results of the full model (see table S2 in the supplement) revealed a similar pattern, but fewer subgroup analyses reached significance, with many p-values narrowly missing the significance threshold. Additionally, a subgroup effect was observed for CS+ and CS- measured with SCR (both $p \leq 0.050$), although both subgroups did not significantly differ from the healthy participants in follow-up analyses. Again, substantial heterogeneity was noted across all analyses (I^2 = 81-87%). These differences in subgroup effects between the full model and the one without outliers indicate that they are more influenced by the exclusion of studies compared to the main findings, warranting greater caution regarding the robustness of these results.

3.1.2 Moderator Analyses: Experimental factors and sample characteristics

Following our criterion of having at least $k \geq 10$ studies for dimensional, and $k \geq 5$ studies for categorical moderator-analyses, we were able to analyze the following moderators in the acquisition phase: reinforcement rate for all outcomes, trait anxiety for all but FPS and depression for both ratings (US expectancy and affective ratings, see supplementary table S3).

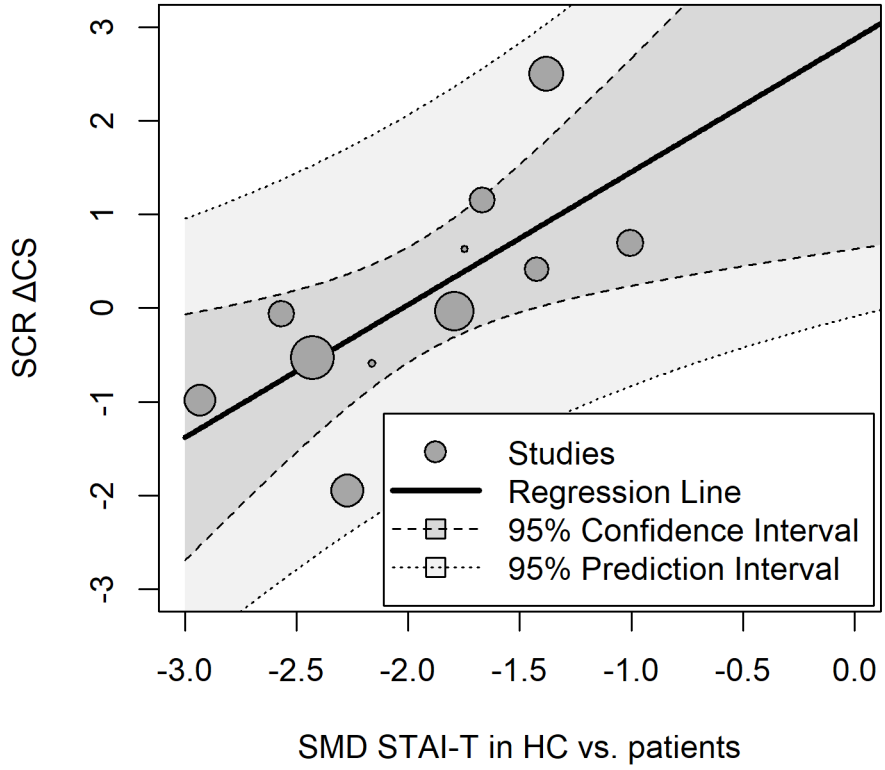
Detailed statistics are available in table S4 in the supplement. Two moderators significantly influenced the results for ΔCS in SCR and FPS. Results indicate that trait anxiety affected responses for ΔCS in SCR ($F(1,9)$ = 8.41, p = .018), specifically showing increasing effect sizes with increasing trait anxiety. The higher the difference of anxiety symptoms between patients and

healthy participants, the smaller the difference between CS+ and CS- (figure 2). Furthermore, reinforcement rate influenced responses for ΔCS in FPS ($F(1,17) = 7.23$, $p = .016$), suggesting larger effect sizes for FPS responses with increasing reinforcement rates. All other moderators examined in the present analysis did not reach significance (all $p > .082$). Overall, the results suggest that variability in outcomes cannot be attributed to the moderating influences of the analyzed variables. Thus, the effect sizes remain consistent across different study characteristics.

As previously described, the outlier analysis led to many subgroup analyses being unfeasible. Therefore, we report in the supplement in Table S5 the full model without excluding outliers. Even here, the number of studies in some cells was too low, but we were able to conduct significantly more analyses. The relationship reported in the corrected model between reinforcement rate and the FPS effect in the ΔCS was replicated; however, the STAI effect for ΔCS SCR response was not replicated ($p = .338$). Additionally, other moderating influences emerged during the acquisition phase. The CS type moderated the effects in the expectancy rating for the CS+, with higher effect sizes observed for pictures and shapes than for the CS-. An influence of the US type on the affect rating of the CS- was also noted; the effect was smaller for visual stimuli and sounds compared to electrotactile stimulation and combinations. This was also reflected in an influence on the ΔCS, which showed less differentiation for these US types. There were no significant findings related to diagnosis-specific influences, which could only be investigated in relation to affect ratings. For the dimensional analyses, additional effects were found concerning the relationship between depressive symptoms and the response to the CS+ in the SCR, as well as the reinforcement rate related to the CS+ SCR response. Increased SCR responses to the CS+ were associated with larger differences in depressive symptoms between the anxiety disorder/OCD group and healthy participant group. A larger SCR response to the CS+ was observed with a higher reinforcement rate, similar to the effect seen with the FPS on ΔCS.

Figure 2

Meta regression examining the impact of the difference in trait anxiety between the anxiety disorder and healthy participant group on SCR to ΔCS in the acquisition phase



Note. Illustration of the result of a meta regression analyzing how differences in trait anxiety (STAI-T) between the anxiety group and healthy participants (HC) affect skin conductance response (SCR) to ΔCS during the acquisition. Larger standardized mean difference (SMD) in STAI-T indicate greater differences for the anxiety patients, zero indicates no group difference and negative values larger effects in healthy participants. SCR = Skin conductance response.

3.2 Extinction phase

In table 5, results of the main model, i.e. the comparison of fear extinction between healthy participants and patients with anxiety- and stress-related disorders for CS+, CS- and ΔCS are shown. Forest plots visualizing the effect sizes of the contributing primary studies and the heterogeneity are presented separately by outcome in supplementary material accompanying this article in figure S2. Compared to healthy participants, patients with anxiety, stress-related disorders and OCD showed increased fear during extinction in affect ratings to both stimuli, i.e., the former threatening stimulus, as well as the safe stimulus (CS+: $g = 0.93$, $p = .024$; CS-: $g = 0.37$, $p < .001$). Regarding the ratings of US expectancy, anxiety patients showed increased fear to the safe

stimulus, i.e. CS-: g = 0.19, p = .001. Focusing on psychophysiological outcomes, i.e. SCR and FPS, we did not find any significant differences between groups. In addition, regarding the ΔCS, there was no significant difference between groups in any of our outcome measurements. In sum, results suggest that patients with stress-related and anxiety-related disorders compared with healthy participants seem to have difficulties in fear extinction learning on a behavioral level as reflected in higher affect ratings to the CS+. Furthermore, they report exaggerated threat (US) expectancy and affect responses to the CS-. The increased responsiveness to the CS+ in affective ratings that was observed during the acquisition phase persists during the extinction phase, just as the safety stimulus CS- also continued to be rated as more aversive in the affect and expectancy rating by patients with anxiety- and stress-related disorders. This might indicate that patients with stress-related and anxiety-related disorders have difficulties to limit their acquired fear to the CS+, which seems to hold at least for the time of extinction training. For the reported models after outlier exclusion, heterogeneity was quite mixed (I^2 = 0-98%).

Results of the full model before outlier exclusion (see table S6 in the supplement) are very similar, as they also show increased responding of patients to the CS- in affect and US expectancy ratings (both $p \leq 0.041$). In addition, in the full model SCRs also indicate increased responding after extinction training to the CS- for patients with anxiety, PTSD and OCD (p = .041). In contrast to the model after outlier exclusion, there is no increased responding in affect ratings to the CS+ of anxiety patients. Importantly, there was again substantial heterogeneity in almost all analyses (I^2 = 52-99%), apart from a few exceptions. Overall, the reported pattern of results is consistent between the model with and without outlier exclusion, which supports the robustness of the analyses.

Table 5

Statistical model parameters of the extinction phase

Outcome measure	Stimulus	Model	Number of studies and participants	*g*	95%CI	*p*	I^2
Skin Conductance Response	CS+	Main model after outlier removal	18 n_{pat} = 433, n_{HC} = 505	0.13	-0.01-0.27	.075	0.0
		Subgroup analysis after outlier removal				.872	
		▪ Anxiety vs. HC	8 n_{pat} = 254, n_{HC} = 267	0.14	0.01-0.27	**.034***	0.0
		▪ PTSD vs. HC	10 n_{pat} = 179, n_{HC} = 238	0.12	-0.18-0.42	.391	0.34

	CS-	Main model after outlier removal	17 n_{pat} = 448, n_{HC} = 508	0.11	-0.07-0.28	.211	0.32
		Subgroup analysis after outlier removal				.275	
		▪ Anxiety vs. HC	9 n_{pat} = 287, n_{HC} = 306	0.04	-0.24-0.32	.751	0.52
		▪ PTSD vs. HC	8 n_{pat} = 161, n_{HC} = 202	0.21	-0.02-0.44	.063	0.0
	ΔCS	Main model after outlier removal	18 n_{pat} = 508, n_{HC} = 537	0.22	-0.92-1.36	.686	0.91
		Subgroup analysis after outlier removal				.331	
		▪ Anxiety disorders and OCD vs. HC	9 n_{pat} = 338, n_{HC} = 317	0.55	-0.08-1.18	.080	0.89
		▪ PTSD vs. HC	9 n_{pat} = 170, n_{HC} = 220	-1.47	-6.22-3.28	.495	0.93
Fear Potentiated Startle	CS+	Main model after outlier removal	8 n_{pat} = 239, n_{HC} = 1138	0.03	-0.55-0.62	.896	0.91
		Subgroup analysis after outlier removal				**.009****	
		▪ Anxiety vs. HC	5 n_{pat} =174, n_{HC} = 1043	-0.29	-1.14-0.56	.403	0.92
		▪ PTSD vs. HC	3 n_{pat} = 65, n_{HC} = 95	0.56	0.08-1.05	**.038***	0.0
	CS-	Main model after outlier removal	5 n_{pat} = 158, n_{HC} = 155	0.08	-0.26-0.42	.540	0.06
		Subgroup analysis after outlier removal				.749	
		▪ Anxiety vs. HC	4 n_{pat} = 137, n_{HC} = 120	0.10	-0.40-0.60	.565	0.28
		▪ PTSD vs. HC	1 n_{pat} = 21, n_{HC} = 35	0	-0.54-0.54	1.000	
	ΔCS	Main model after outlier removal	5 n_{pat} = 158, n_{HC} = 155	0.19	-0.53-0.91	.511	0.75
		Subgroup analysis after outlier removal				**<.001****	
		▪ Anxiety disorders and OCD vs. HC	4 n_{pat} = 137, n_{HC} = 120	-0.04	-0.32-0.24	.674	0.0
		▪ PTSD vs. HC	1 n_{pat} = 21, n_{HC} = 35	1.21	0.62-1.80	**<.001****	
US Expectan	CS+	Main model after outlier removal	17	0.56	-0.36-1.47	.214	0.0

			n_{pat} =383, n_{HC} = 2214				
		Subgroup analysis after outlier removal				.231	
		▪ Anxiety vs. HC	4 n_{pat} = 144, n_{HC} = 1004	-0.09	-1.56-1.41	.854	0.95
		▪ PTSD vs. HC	13 n_{pat} = 238, n_{HC} = 1210	0.77	-0.42-1.95	.183	0.98
	CS-	Main model after outlier removal	12 n_{pat} = 255, n_{HC} = 298	0.19	0.09-0.29	**.001****	0.0
		Subgroup analysis after outlier removal				.415	
		▪ Anxiety vs. HC	3 n_{pat} = 107, n_{HC} = 81	0.20	-0.32-0.71	.243	0.0
		▪ PTSD vs. HC	9 n_{pat} = 148, n_{HC} = 217	0.18	0.08-0.29	**.004****	0.0
	ΔCS	Main model after outlier removal	15 n_{pat} = 383, n_{HC} = 2214	0.06	-0.002-0.13	.059	0.0
		Subgroup analysis after outlier removal				.231	
		▪ Anxiety disorders and OCD vs. HC	3 n_{pat} = 144, n_{HC} = 1004	0.06	-0.47-0.58	.694	0.0
		▪ PTSD vs. HC	12 n_{pat} = 239, n_{HC} = 1210	0.07	0.01-0.12	**.026***	0.0
Affect Ratings	CS+	Main model after outlier removal	27 n_{pat} = 466, n_{HC} = 1488	0.93	0.13-1.73	**.024***	0.98
		Subgroup analysis after outlier removal				.871	
		▪ Anxiety vs. HC	10 n_{pat} = 262, n_{HC} = 286	0.25	-0.27-0.77	.307	0.83
		▪ PTSD vs. HC	17 n_{pat} = 204, n_{HC} = 1202	1.933	0.08-2.57	**.038***	0.98
	CS-	Main model after outlier removal	22 n_{pat} = 277, n_{HC} = 430	0.37	0.19-0.56	**<.001*****	0.19
		Subgroup analysis after outlier removal				.206	
		▪ Anxiety vs. HC	6 n_{pat} =115, n_{HC} = 151	0.21	-0.21-0.64	.247	0.37
		▪ PTSD vs. HC	16 n_{pat} = 162, n_{HC} = 279	0.46	0.24-0.68	**<.001*****	0.06
	ΔCS	Main model after outlier removal	26 n_{pat} =, n_{HC} =	0.02	-0.14-0.18	.780	0.21

		Subgroup analysis after outlier removal			.531	
		▪ Anxiety disorders and OCD vs. HC	10 n_{pat} = 262, n_{HC} = 286	0.13	-0.30-0.56	.508	0.64
		▪ PTSD vs. HC	16 n_{pat} = 162, n_{HC} = 279	0.005	-0.14-0.15	.944	0.0

Note: n_{pat} = sample size patients, n_{HC} = sample size healthy participants, g = Hedge's g, CI = confidence interval, PTSD = Posttraumatic stress disorder, Anxiety = All anxiety disorders except for PTSD. Text written in grey: subgroup specific effects when subgroup analyses were non-significant*$p < .05$, **$p < .01$, ***$p < .001$.

3.2.1 Moderator Analyses: Subgroup analyses PTSD vs. other anxiety disorders

During the extinction phase, subgroup analyses suggest a significant difference in FPS responses to the CS+ and ΔCS between PTSD patients and those with anxiety disorders and OCD (both $p \leq$ 0.009). For both stimuli CS+ and ΔCS, we obtained a significant subgroup-specific effect for the PTSD group (CS+: $g = 0.56$, $p = .038$, n = 3; ΔCS: $g = 1.21$, $p < .001$, n = 1), which was absent for anxiety and OCD. So while no overall effects were observed for FPS, PTSD patients showed elevated FPS to the CS+ and ΔCS. Please note the limited number of studies included. Also, note that other subgroup effects did not reach statistical significance. For completeness, the table includes follow-up analyses for both groups separately in these instances as well; however, we refrain from interpreting these results. The estimated between-study heterogeneity varied strongly in both groups (I^2 = 0- 98%), suggesting large variety between studies (table 5 depicts detailed statistics).

Analyses including outliers (see table S6 in the supplement) show a similar pattern of results, suggesting that PTSD patients show larger ΔCS responses in FPS, however not to the CS+. In addition, we observed a stronger responding to the CS- in SCR in PTSD.

3.2.2 Moderator Analyses: Experimental factors and sample characteristics

The following moderators were analyzed during extinction in the model after outlier exclusion: reinforcement rate for US expectancy and affective ratings (see supplementary table S7) and depression score for affective ratings. In sum, we did not find any significant effect of any of the moderators. In contrast to the reduced model, the full model without outlier exclusion revealed several significant moderating influences, as detailed in Table S9 of the supplement. Similar to the acquisition phase, both CS and US type moderated the findings, with stronger effects observed for pictures as the CS type in the SCR response to CS+. The US type also moderated the CS+ response, with stronger effects noted for electrotactile stimulations in the affect rating. No effects were found for diagnosis-specific influences, which could only be analyzed in relation to affect ratings.

Regarding dimensional predictors, an influence of reinforcement rate on the CS- SCR response was observed, with higher responses during extinction being associated with a higher reinforcement rate during acquisition. Additionally, depressive symptoms influenced affect ratings for CS- and the ΔCS. Increased affect ratings for CS- were associated with larger differences in depression scores between the anxiety/OCD groups and healthy participant groups. In contrast, larger differences in depression scores between the anxiety/OCD groups and healthy participant groups were associated with smaller ΔCS affect ratings (Table S9), possibly explained by a descriptive influence of increased depression score differences on smaller CS+ affect ratings, which was beyond the significance threshold.

3.3 Extinction recall phase

In contrast to fear acquisition and fear extinction, fear extinction recall has only been investigated in a limited number of studies (see table 6). In line with the other phases, we also performed outlier analyses and report results after those analyses. However, outliers were only detected for SCRs. Therefore, we created only one table, including the results with and without outliers. Forest plots visualizing the effect sizes of the contributing primary studies and the heterogeneity are presented separately by outcome for CS+ and CS-, and ΔCS in supplementary material accompanying this article in figure S5. Patients with anxiety- and stress-related disorders seem to have ongoing aberrancies in fear extinction learning in subjective ratings as indicated by increased ratings to the CS- (US expectancy: g = 0.37, 95% CI: 0.05-0.69; p = .030, affect rating: g = 0.30, 95% CI: 0.11-0.50; p = .007) compared with healthy participants during extinction recall. In addition, again increased affect ratings for the CS+ were observed in patients (g = 0.46, 95% CI: -0.21-0.72; p = .003). Regarding the SCR, results indicate increased responding to the CS+ in patients with anxiety- and stress-related disorders in extinction recall (g = 0.28, 95% CI: 0.01-0.55; p = .044). In contrast, patients with anxiety disorders, PTSD and OCD do not seem to show exaggerated fear, as measured via FPS. Results even suggest increased responding for healthy participants to the CS- during extinction recall (g = -0.21, 95% CI: -0.23-(-0.20); p = .004). However, since only two studies contributed to this finding, the interpretative power is limited, and we will not discuss the effect in depth. Overall, analyses of the extinction recall phase mirror results of the extinction phase, indicating persistent difficulties in learning the new safety association, effect sizes are overall small, and only the increased responding to the CS+ in affective ratings was of medium effect size (Hedges, 1981). Between-study heterogeneity was rather low (I^2 = 0-47%), indicating the reliability of these results (Ryan, 2016). Due to the limited number of studies, we were not able to conduct subgroup analyses for the extinction recall phase. A graphical representation of the effect sizes, as

well as a summary of the results across all phases (i.e, acquisition, extinction, and extinction recall), separately for different outcome measures and CS+. CS- and ΔCS can be found in figures 3 and 4, respectively.

Table 6

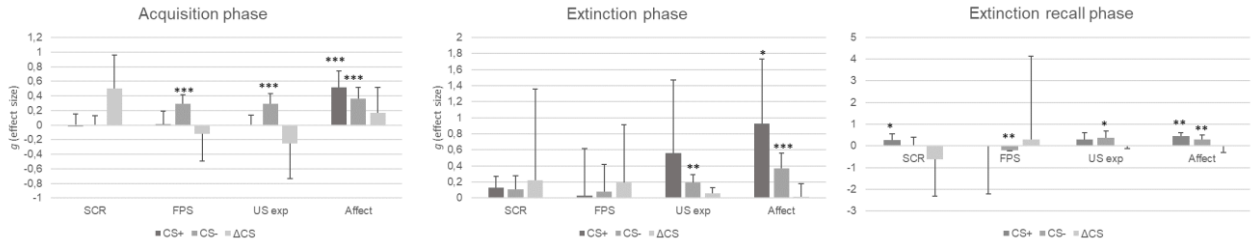
Statistical model parameters of the extinction recall phase

Outcome measure	Stimulus	Number of studies and participants	*g*	95%CI	*p*	I^2
Skin Conductance Response	CS+	15 n_{pat} = 348, n_{HC} = 291	0.28	0.01-0.55	**.044***	0.47
	CS-	11 n_{pat} = 261, n_{HC} = 231	-0.04	-0.34-0.26	.771	0.43
	CS- (with outlier)	12 n_{pat} = 265, n_{HC} = 245	0.05	-0.36-0.46	.797	0.60
	ΔCS	10 n_{pat} = 241, n_{HC} = 206	0.60	-1.14-2.34	.455	.92
	ΔCS (with outlier)	13 n_{pat} = 268, n_{HC} = 248	-0.61	-14.93-13.71	.928	.95
Fear Potentiated Startle	CS+	2 n_{pat} = 94, n_{HC} = 70	-0.003	-2.23-2.22	.990	0.10
	CS-	2 n_{pat} = 94, n_{HC} = 70	-0.21	-0.23-(-0.20)	**.004****	0.0
	ΔCS	2 n_{pat} = 94, n_{HC} = 70	0.30	-3.53-4.13	.504	.67
US Expectancy Rating	CS+	8 n_{pat} = 153, n_{HC} = 161	0.29	-0.03-0.62	.071	0.17
	CS-	8 n_{pat} = 153, n_{HC} = 161	0.37	0.05-0.69	**.030***	0.19
	ΔCS	8 n_{pat} = 153, n_{HC} = 161	-0.03	-0.11-0.06	.450	0.0
Affect Rating	CS+	10 n_{pat} = 80, n_{HC} = 142	0.46	0.21-0.72	**.003****	0.0
	CS-	10 n_{pat} = 80, n_{HC} = 142	0.30	0.11-0.50	**.007****	0.0
	ΔCS	10 n_{pat} = 80, n_{HC} = 142	-0.01	-0.30-0.28	.940	0.0

Note: n_{pat} = sample size patients, n_{HC} = sample size healthy participants, g = Hedge's g, CI = confidence interval, *p < .05, **p < .01.

Figure 3

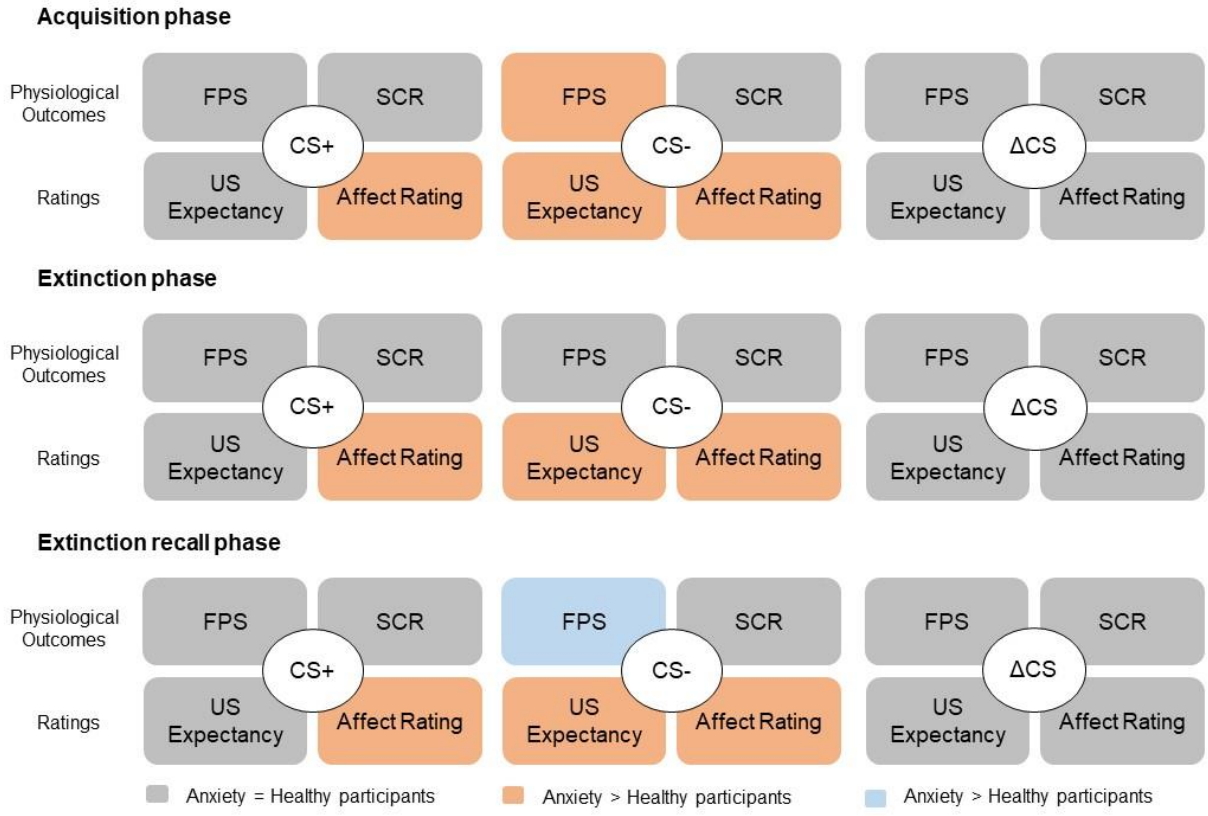
Graphical representation of the effect sizes per stimulus type in anxiety patients vs. healthy participants.



Note: Bars reflect the standardized mean difference of the model after outlier removal in fear responses between anxiety patients minus healthy participants. Hence, positive effect sizes indicate stronger responses by anxiety patients compared to healthy participants. Error bars display the 95% confidence interval. *p < .05, **p < .01, ***p < .001.

Figure 4

Visualization of results across the acquisition, extinction and extinction recall phase for the different stimuli and outcomes



Note: Four outcome measurements (FPS, SCR, US expectancy and affect rating) have been used to examine responding to the conditioned stimuli, CS+, CS-, and ΔCS. Grey color indicates no group difference, orange colored rectangles represent those

outcome measurements in which patients with an anxiety disorder showed an increased response compared with healthy participants, and blue color indicates increased responses for healthy participants compared to patients with an anxiety- or stress-related disorder. . FPS = Fear potentiated startle; SCR = Skin conductance response; GAD = Generalized anxiety disorder, SAD = Social anxiety disorder; PTSD = Posttraumatic stress disorder.

3.4 Publication bias

To investigate publication bias, we visually inspected the funnel plots of the effect sizes of all outcome measurements for both stimuli (see figure S3 and S4 in the supplement). In addition, we performed an Egger's test to test for asymmetry in the funnel plot. For the acquisition phase, the Egger's intercept of the funnel plot was significant for ΔCS measured with SCR (B = 4.91, p = .044), and US expectancy rating (B = -4.52, p = .005), indicating evidence of publication bias. For the extinction phase, publication bias is indicated for the distribution of US expectancy effect sizes for the CS+ (B = 9.66, p = .004), and the CS- (B = 0.82, p = .004). In these cases, it must be considered that the effect sizes reported here are likely overestimated due to publication bias. In addition, we performed *p*-curve analyses. Because of the heterogeneity in our data, it was only possible to conduct those analyses for some but not all outcome measures. Therefore, we refer interested readers to the results of the analyses in the supplement (figure S3 and S4).

4 Discussion

Considering the significance attributed to fear and safety learning in both pathophysiological models and therapeutic strategies for anxiety and related disorders (Kitt et al., 2023; Mineka & Oehlberg, 2008; Pittig et al., 2018), our objective was to integrate the outcomes of fear conditioning studies investigating fear and safety learning in individuals with anxiety disorders, OCD and PTSD. Building upon the meta-analysis by Duits et al. (2015), we integrated findings across anxiety- and stress-related disorders, and we conducted additional analyses separately for PTSD vs. anxiety disorders and OCD. Furthermore, we integrated results of different outcome measurements separately for affect ratings, expectancy ratings, FPS and SCR. In total, 77 fear conditioning studies are included in the present meta-analysis, comprising n = 5310 individuals (2052 patients with anxiety disorders, OCD, or PTSD, and 3258 healthy participants). Key findings are:

(1) During fear acquisition, when participants initially learn about the association between a cue and a threat, no significant differences in the physiological responses or expectancy ratings to the CS+ between patients with stress - and anxiety-related disorders and healthy participants emerge. However, patients with stress- and anxiety-related disorders tend to affectively rate the threatening stimulus CS+ as more aversive, an effect driven by patients with anxiety and OCD

specifically. Simultaneously, in the context of fear acquisition, the CS- (a cue that was never paired with a US) leads to an elevated FPS, increased threat expectancy, and greater affect ratings in individuals with stress- and anxiety-related disorders compared to healthy participants. Note that the effects sizes for the observed group differences during acquisition are small (US-expectancy rating, FPS) to moderate (affect ratings). When comparing patients with PTSD vs. anxiety and OCD most effects hold for both clinical groups, even though the effects were larger for anxiety/OCD. Further, only anxiety patients and OCD but not PTSD patients show an elevated SCR response to the ΔCS as well as elevated affect ratings to the CS+ (large effect sizes).

(2) During fear extinction, after the formerly threatening stimulus (CS+) is no longer paired with a threat, individuals with anxiety- and stress-related disorders continue to show heightened threat expectancy towards the CS- (small effect size), and increased affect ratings to both the CS- (medium effect size) and the CS+ (large effect size) compared to healthy participants. In addition to the effects observed for both patients with anxiety/OCD and PTSD, PTSD patients show larger FPS responses to the CS+ and ΔCS.

(3) During extinction recall, that examines the persistence of extinction learning, the results parallel those of the extinction phase, showing higher ratings of affect and expectancy for the CS- together with increased affect ratings to the CS+ in patients with anxiety- and stress-related disorders compared to healthy participants (medium effect sizes).

(4) In general, the observed heterogeneity was often considerable; however, the number of studies was frequently insufficient for more in-depth moderator analyses. The reinforcement rate and trait anxiety symptoms moderated responding to the ΔCS during the acquisition phase. Specifically, we found that both the presence of greater differences in anxiety symptoms between groups, as well as a higher reinforcement rate was associated with a smaller differentiation of CS+ and CS- (measured with SCR and FPS, respectively). The full model including the outliers revealed additional moderating influences related to US type, CS type, and depressive symptoms during both acquisition and extinction. Furthermore, it is essential to highlight that the analyses indicated a risk of publication bias, which could lead to distortions and overestimations of the findings—particularly concerning affect and expectancy ratings during acquisition, as well as SCR and US expectancy during extinction.

In the following sections, we will discuss central findings of our meta-analysis, including the heightened reactivity of anxiety patients to the safety cue CS- during acquisition and extinction, along with their sustained exaggerated response to the CS+ during acquisition, extinction and extinction recall in affective rating results. The heightened responses to a safe stimulus (CS-) that has not been directly associated with an aversive outcome, observed at both physiological and self-

report levels during acquisition and extinction, not only replicates but also strengthens findings from prior studies (Craske et al., 2012; Lissek et al., 2009) and the meta-analysis by Duits et al. (2015). Effects sizes of the increased responding to the CS- during acquisition were small to medium (FPS and US expectancy ratings: g = 0.29; affect rating: g = 0.36) and comparable to those reported by Duits et al. (d = 0.30). Importantly, this effect persists during extinction in which patients with stress-related and anxiety disorders continue to show increased responding towards the CS- in affect and expectancy ratings in our study (US expectancy ratings: g = 0.19; affect rating: g = 0.37), an effect similarly seen in Duits et al for subjective ratings (d = 0.26). Our analyses extend these findings and demonstrate that, even in extinction recall, patients with anxiety-related and stress-related disorders subjectively evaluate the CS- as more threatening (US expectancy ratings: g = 0.37; affect rating: g = 0.30). Several underlying mechanisms of this phenomenon have been discussed including impaired safety learning, deficient fear inhibition to stimuli signaling safety, or fear overgeneralization (Beckers et al., 2023; Duits et al., 2015; Jovanovic et al., 2012; Lissek et al., 2005). Overall, the findings of this meta-analysis indicate that heightened responses to the safe cue are present across various stress-related and anxiety disorders, including OCD, and remain stable throughout different phases of fear conditioning experiments, persisting even during extinction. For patients with stress-related and anxiety-related disorders, this results in a diminished ability to recognize safety cues, which can lead to increased uncertainty, stress, arousal, and anxiety, which could contribute to heightened levels of distress (Staples-Bradley et al., 2018) or perception of generalized unsafety (Brosschot et al., 2018). In a broader context, difficulties in discriminating between threat and safety signals are in line with predictive processing frameworks, which posit a common *"better-safe-than-sorry"* processing strategy as a general vulnerability factor for anxiety or negative affect more broadly (Brosschot et al., 2018; de Jong & Vroling, 2013; Van den Bergh et al., 2021). More specifically, this refers to an information processing heuristic characterized by reduced sensory-perceptual detail, allowing threat-related priors to dominate experience, thereby enhancing the detection of potential threats at the expense of precision (Van den Bergh et al., 2021). In the context of fear conditioning, this phenomenon would manifest as heightened threat responses to stimuli that are, in fact, safe, as indicated by this meta-analysis. While such a processing mode may be beneficial in threatening environments, it can lead to difficulties in identifying safe stimuli and inhibiting the fear response. Furthermore, whenever a threat is detected, it triggers an immediate, reflexive, associative reaction, followed by a more effortful and deliberate cognitive evaluation process (Buhle et al., 2014; Ouimet et al., 2009; Wen et al., 2024). In conjunction with reduced precision and consequently poorer stimulus discrimination in anxiety—which may increase sensitivity to detecting threats—it appears that the subsequent,

more deliberate top-down control over conditioned responses and the inhibition of cued fear, mediated by the medial prefrontal cortex (mPFC), are also compromised in individuals with anxiety (Sangha et al., 2020). Difficulties in detecting safety stimuli, which contribute to heightened fear responses, combined with impaired ability to inhibit these responses in anxiety-related and stress-related disorders, may exacerbate symptoms and sustain anxiety in stress- and anxiety-related disorders.

In addition to the aforementioned anomalies in responses to the safety stimulus during acquisition, extinction, and extinction recall, there is also an enhanced aversive evaluation of the CS+ during acquisition, extinction and extinction recall in self-reported affect. These findings align well with and extend the previous findings by Duits and colleagues that report an increased responding to the CS+ for the extinction phase only. Note that effects sizes are larger in our study and of medium - large magnitude (Duits et al, extinction: $d = 0.35$, our study affect ratings during acquisition: $g = 0.52$, extinction: $g = 0.93$, extinction recall: $g = 0.46$) (Duits et al., 2015). Overall, this supports that individuals with anxiety and stress-related disorders show heightened affective responses to a threatening stimulus (CS+) throughout the various phases of fear conditioning. Beginning with the acquisition phase, when the fear association is being formed, individuals with anxiety and stress-related disorders report higher affective ratings for the CS+. This finding remains stable even during extinction and extinction recall, despite the CS+ no longer being paired with a threat. This heightened response during extinction has been related to slower extinction processes in anxiety- and stress-related disorder that may contribute to the maintenance and exacerbation of anxiety disorders (Maren & Holmes, 2016; Milad & Quirk, 2012). The inhibitory retrieval model (Craske et al., 2014) postulates that during extinction, the original fear memory remains intact while a new inhibitory memory trace is formed (Bouton, 2004, 2014; Bouton et al., 2021), and that this inhibitory process is disrupted in anxiety which leads to a dominance of the original fear memory trace. Extinction seems to rely on activity in a neural network centered on the amygdala, the vmPFC, and the hippocampus with a central role of plasticity in the infralimbic cortex as a region within the vmPFC which is important for inhibiting subcortical structures and emotional responses such as fear (Sangha et al., 2020). The inhibitory retrieval model has led to relevant clinical implications for exposure therapy, as reviewed by Craske and colleagues, such as optimizing therapy outcomes by increasing prediction error through expectancy violation and enhancing consolidation (Craske et al., 2022). However, in addition to a fear inhibitory mechanism, other mechanisms and interventions may also be of interest to suppress learned fear. Research conducted in rodent models indicate that the mechanisms involved in fear extinction can change depending on the frequency and duration of extinction training. Initially, inhibitory processes appear to be pivotal,

with erasure mechanisms becoming more prominent and taking control as the extinction training proceeds (An et al., 2017). This emphasizes the importance of the duration and repetition of exposures in the treatment of anxiety disorders to enhance and sustain the efficacy of exposure therapy. Moreover, targeting multiple learning mechanisms related to extinction learning along with learning across diverse contexts and situations may lead to more stable learning outcomes (Gershman et al., 2010). In addition, several further manipulations have been explored to enhance extinction training, such as pharmacological interventions to disrupt reconsolidation (Kindt et al., 2009), or enhance extinction (Stafford et al., 2012). We focus on extinction learning and the translational insights that can be derived from laboratory studies in this research field for exposure-based interventions. Although this is a significant field of research, it is important to acknowledge that extinction, while presenting an interesting and plausible model for exposure-based therapies, has its limitations (Scheveneels et al., 2016). For example, the types of stimuli and outcome measures differ considerably from those used in clinical exposure procedures, as do the timing factors (e.g., single sessions versus extended learning experiences over several weeks; Kredlow et al., 2022). Moreover, several mechanisms that contribute to the success of exposure therapies are often overlooked in extinction models, including instrumental learning, actively approaching aversive situations, avoidance, habituation, and cognitive factors such as expectation violation (see Benito et al., 2024 and Knowles & Tolin, 2022 for detailed reviews). Future research directions that aim to enhance insights for exposure-based therapies may benefit from utilizing more ecologically valid and complex stimuli, such as the use of virtual reality (Kredlow et al., 2022; Scheveneels et al., 2016). Additionally, it is essential to acknowledge and integrate other relevant mechanisms and their interactions. For instance, sustained fear responses during extinction promote avoidance behavior (Cain, 2019; de Haart et al., 2021; Hunt et al., 2022) which is known to contribute to the maintenance of anxiety and to hinder extinction (Kausche et al., 2022; Pittig et al., 2018). This reinforces and strengthens anxiety-related patterns, potentially expediting the chronicity and increased severity of anxiety disorders. Shedding light on the interplay between fear learning, extinction, and other relevant psychopathological mechanisms such as avoidance is crucial to comprehend the complex processes involved in the development and maintenance of anxiety disorders, to inspire the search for effective interventions.

It is noteworthy that we observed increased responding to both CS+ and CS- across all phases of fear conditioning, resulting in a similar level of discrimination between CS+ and CS- in healthy controls and patients with stress- and anxiety-related disorders. This suggests an overall heightened reactivity to safe and threat cues rather than improved differentiation between safe and threatening cues. This effect may be attributed to generalization (Cooper et al., 2022), general

hyperarousal (Grillon, 2008) or increased sensitivity to threat (Naragon-Gainey, 2010), functioning as a cognitive bias in stress-related and anxiety disorders, resulting in exaggerated responses to both threatening and safe stimuli (Chaaya et al., 2018). This heightened reactivity limits the ability to differentiate between CS+ and CS-, reducing the distinction between safe and threatening cues. Deficits in hippocampus-mediated pattern separation, a process through which memories are encoded as distinct representations, have been proposed as a potential underlying marker for difficulties in differentiation and increased generalization in fear conditioning (Lopresto et al., 2016). Additionally, the emotional context during conditioning may further influence responses to both CS+ and CS-, explaining the exaggerated reactions (Andreatta et al., 2020) and generalization. In summary, the lack of increased differentiation between CS+ and CS- in anxiety conditioning studies can reflect a broader pattern of heightened reactivity and difficulty in distinguishing between safe and threatening stimuli or context generalization effects. Furthermore, it highlights the possibility that relying solely on difference measures in conditioning studies may obscure a general alteration in response tendencies.

While the subgroup analyses generally support the reported pattern of results for both patients with PTSD and those with anxiety disorders and OCD, certain differential effects must be noted. Specifically, the observed effects were significantly stronger in individuals with anxiety disorders and OCD during the acquisition phase. Follow-up subgroup analyses indicate that the heightened affective ratings in response to the CS+ during acquisition are specific to anxiety disorders and OCD. Furthermore, only patients with anxiety disorders and OCD exhibited heightened SCR to the ΔCS, while patients with PTSD did not show these effects. Additionally, effect sizes for stronger CS- responding during acquisition in the PTSD group were small (US expectancy $g = 0.12$; affect rating $g = 0.17$), whereas effect sizes were significantly higher and of medium size for anxiety disorders and OCD (US expectancy $g = 0.58$; affect rating $g = 0.47$). In contrast, during the extinction phase, larger effects were observed in PTSD patients, with alterations in fear-potentiated startle—specifically, a stronger startle response to the CS+ and ΔCS manifesting only in patients with PTSD, revealing medium to strong effect sizes. These effects were not apparent in patients with anxiety disorders or OCD. Overall, the pattern of results suggests that there are more alterations during the acquisition phase in anxiety patients and OCD, while stronger effects appear during extinction in PTSD patients. However, it remains unclear whether this finding is replicable. Notably, this differential effect was not reported in the analyses conducted by Duits et al., underscoring the need for future studies to further investigate this effect, its robustness, and its implications for treatment. Phenomenologically, it may align with the notion that the symptomatology in PTSD typically begins after an extremely intense learning event and life

stressor, such as a trauma, which would be significant for everyone. In contrast, anxiety patients often seem to lack clear initial learning experiences, which may be related to heightened sensitivity and altered processes in the acquisition of fear. These findings may contribute to the existing literature that emphasizes both shared and distinct psychopathological processes in fear conditioning between PTSD and anxiety disorders (Acheson et al., 2015), as well as differences in brain structure (Radua et al., 2010), and etiological factors (Chantarujikapong et al., 2001). Overall, there are only a few studies that directly compare different disorders with each other, and more studies in this area would be desirable. Similarly, a network meta-analysis that enables direct and indirect disorder comparisons could enhance future insights into group comparisons. Given the current data situation, this would only be possible through aggregation across different outcomes.

Even after considering subgroups, there remains a high degree of heterogeneity in the findings. Subgroup analyses in the full dataset including outliers support the notion that some of this heterogeneity is likely attributable to methodological factors, such as US type, CS type, and reinforcement rate—all of which have previously been linked to differences in fear conditioning (e.g., Ahs et al., 2018; Lonsdorf et al., 2017; Ney et al., 2022). In addition, participant diversity, symptom variability, and reliance on diagnostic categories that exhibit high comorbidity and heterogeneity likely also contribute to the overall heterogeneity. Dimensional analyses regarding differences in trait anxiety and depressive symptoms between clinical groups and healthy participants in primary studies in this meta-analysis suggest that higher symptom severity in trait anxiety was associated with larger SCR responses to the ΔCS during acquisition. Note, that the effects of differences in depressive symptoms were observed in the full model, including outliers, linking depressive symptoms to changes in fear learning processes (Luo et al., 2024; Otto et al., 2014). More precisely, without excluding outliers, we found that larger differences in depressive symptoms between the clinical groups and healthy participant groups were linked to decreased ΔCS in affect ratings, which was driven by increased affect ratings to the CS- and descriptively decreased affect ratings to the CS+. In the acquisition phase, the full models including outliers indicated that increased depressive symptoms were associated with increased SCR to CS+. Of note, only a subset of studies reported differences in depressive symptoms, thus this pattern of results requires further investigation. Collectively, the results indicate that differences in depressive symptoms between groups might modulate SCR during acquisition and affect ratings during extinction, highlighting the need to move beyond comparing different disorder categories and consider the complex heterogeneous nature of mental disorders and their co-occurrence using dimensional analyses. However, due to heterogeneity in reported clinical measures in primary studies, our data did not sufficiently allow for more detailed analyses of specific transdiagnostic

symptom dimension associations. Therefore, we cannot conclusively determine whether the observed patterns of results are specifically related to certain symptom profiles or more broadly linked to anxiety. Furthermore, our results have been identified in the context of heterogeneous often multisymptomatic patients having comorbid diagnoses, thus it appears promising to reclassify patients based on their dysfunctions in fear conditioning and extinction (Fyer et al., 2020). The exploration of data-driven subtypes has the potential to elucidate associations with symptomatology at a more granular level and simplify the complexity of analyses by providing clearly defined subtypes to delve deeper into the neurobiological foundations of these dysfunctions. Furthermore, it offers the opportunity to personalize treatment based on observed aberrancies in fear learning processes, e.g., focusing on improving discriminating between threat and safety signals or enhancing extinction learning. This approach may ultimately inform the development of more effective and personalized treatment strategies.

Overall, the analysis reveals more effects in the realm of subjective ratings, particularly in affective evaluations, while physiological effects were only seen for FPS responses to the CS- during acquisition, as well as exaggerated FPS responding to CS+ and ΔCS in PTSD. This underscores the distinction between different anxiety systems and the discrepancy between subjective ratings and physiological responses. In the context of our data we observed most robust differences on a self-report level, and only robust alterations in fear potentiated startle during CS- responding during acquisition. The two-system framework by LeDoux and Pine (2016) can be used to explain these differences. Based on this theory, the used outcome measurements may reflect different processes of fear learning. The framework distinguishes between two processes involving distinct neural regions: the conscious experience of "fear," mainly mediated by cortical areas, and the automatic behavioral and physiological responses to threats, predominantly controlled by subcortical regions such as the amygdala. This differentiation is supported by a meta-analysis showing distinct neural mechanisms underlying explicit (involving the pulvinar and parahippocampal gyrus) and implicit fear processing (involving a cerebellum-amygdala-cortical pathway) (Tao et al., 2021). Furthermore, a recent study examining the correspondence of physiological activity and arousal and valence rating using representational similarity analysis (RSA) supports that similar affective experiences are related to distinct physiological responses, emphasizing divergence of physiological and self-report data and the importance of considering intra-individual variability (Koppold et al., 2024). Notably, no robust group differences were observed for SCR, which are among of the most commonly used outcome measures in fear conditioning studies. This finding contributes to existing research highlighting the high

methodological heterogeneity in SCR studies (Sjouwerman et al., 2022), which may limit both reliability and validity.

In summary, this meta-analysis strengthens and expands the central findings of Duits by revealing heightened responses especially to safety cues during initial learning, as well as its persistence after extinction in stress- and anxiety-related disorders. Conceptually, it is evident that alterations in fear and safety learning are frequently regarded as risk or vulnerability markers in research studies and psychopathological models, contributing to the onset and persistence of stress- and anxiety-related disorders. These are seen as enduring traits of an individual that predate the disorder and contribute to its emergence, as well as predicting the course of the disorder and response to treatment. However, most studies conducted thus far are cross-sectional, which limits conclusions about causality and predictability. Thus, differences in acquisition/extinction/extinction recall may either be a consequence of anxiety and stress symptomatology, a risk factor, or both. Furthermore, several important prerequisites for a marker to be considered a reliable vulnerability indicator or endophenotype have not been tested (Gottesman & Goud, 2003; Miller & Rockstroh, 2013). It is essential to demonstrate that alterations are present before the onset of the disorder and in individuals at increased risk for developing the disorder, such as in unaffected first-degree relatives. To the best of our knowledge, this has not yet been tested in the field of fear conditioning. Furthermore, a vulnerability marker should possess predictive validity to forecast the course of symptoms. There are few studies examining the predictive validity for the onset of symptoms and treatment trajectories. A longitudinal investigation that supports predictive validity showed that a weaker neural discrimination between the CS+ and CS- in vmPFC during fear acquisition was associated with a greater increase in fear symptoms over the next 2.5 years (Peng et al., 2023). Further, as reviewed by Scheveneels et al. results suggest that reduced extinction and broader generalization predict higher anxiety levels (Scheveneels et al., 2021). In relation to the predictive validity of alterations in fear conditioning for treatment outcome, only a very limited number of studies did examine this with some studies do find an association with extinction and therapy outcome whereas others do not (Scheveneels et al., 2021). Several studies have established a connection between changes in fear conditioning and therapeutic success, especially within the framework of exposure therapy. In the case of spider phobia, an association was shown between pre-treatment differences in fear ratings for CS+ and CS- and symptom reductions. Further, differences in extinction learning in OCD were linked to CBT outcome (Geller et al., 2019). A twin study on monozygotic twins discordant for combat exposure and PTSD suggests that deficient retention of extinction in conditioned fear occurs only in those with PTSD not their co-twins, supporting the conclusion that this deficit is a result of combat trauma leading to PTSD, rather than

a predisposing factor shared within families (Milad et al., 2008). This highlights environmental factors and learning experiences as sources of variation in fear learning, which is further supported by a recent meta-analysis suggesting that adverse childhood experiences lead to diminished CS+/CS- discrimination during acquisition and generalization, driven by decreased responses to CS+ (Ruge et al., 2024). In summary, the empirical basis for evaluating changes in fear conditioning as vulnerability markers for anxiety, PTSD, and OCD is insufficient and does not support this far-reaching perspective. Future studies incorporating high-risk studies and longitudinal designs, along with a more cautious approach to assumptions of causality, are both necessary and important.

4.1 Limitations and recommendation for future studies

There are some limitations to our meta-analysis. First, the study selection may have introduced a bias, as only published studies were included and several studies had to be excluded due to insufficient details for calculating effect sizes. Second, we included only studies that used a patient sample based on a clinical interview. However, there was no information of clinical expertise, why there is the possibility of insufficient clinical expertise in the diagnosis depending on the clinical training as a potential limitation of the validity of disorder diagnosis. Third, there are some indications of publication bias, which may have led to an overestimation of effect sizes and biases in the reporting of results, and which may have even underestimated the effects since only published studies were considered. Although, results of our Egger's test and *p*-curve analyses indicate different fields for publication bias, which limits interpretation of publication bias. Fourth, the heterogeneity was substantial, and the number of outliers was relatively high. While their exclusion in our main models increased robustness, it may have introduced bias due to the removal of several studies. Therefore, we also reported the model without outlier removal in the supplementary materials and highlighted discrepancies in the main paper. Additionally, many influences in the moderator analyses could not be computed due to the limited number of studies remaining for specific moderators, especially after outlier exclusion. Once again, the effects of outlier exclusion are detailed in the supplement and referenced in the paper. In this context, we would like to refer once again to the information in our tables 2 and 3, in which the heterogeneity between the studies and the difficulty of reconciling them becomes apparent. Our meta-analysis highlights the substantial heterogeneity across studies, encompassing variations in study design, stimulus type (CS and US), the number of stimulus presentations. In addition, reading through the primary studies, it becomes apparent that even if the same outcome measurements were used, their application and processing differed between studies. These differences pose a significant challenge

to synthesizing results and deriving clear conclusions, particularly when aiming to explore nuanced aspects such as the characteristics of different anxiety disorders. This between-study heterogeneity underscores the pressing need for greater standardization in fear conditioning research. Future studies should strive to align methodologies, including standardized protocols for stimulus presentation, measurement, and analysis, to facilitate more meaningful comparisons and advance our understanding of the mechanisms underlying anxiety disorders. Without such harmonization, the interpretability and applicability of meta-analytic findings remain limited, potentially hindering progress in the field. Also, only some studies provided information about medication. So we could not include this as a possible moderator in our analyses, why we cannot rule out the possibility of medication influencing our results. Moreover, there are additional variables (e.g., specific symptom dimensions of anxiety, symptom severity, and assessment of functioning) that could explain variance across studies but were not systematically assessed in a similar manner across studies and therefore could not be examined. This highlights the need to define and adhere to standards in studying fear conditioning (Bach et al., 2023; Lonsdorf et al., 2017). Moreover, investigating fear learning processes on a trial-by-trial basis would give the possibility to investigate a more dynamic aspect of fear learning. Furthermore, the study focuses on laboratory studies of fear conditioning using aversive stimulations like shocks or sounds, limiting the ecological validity of studies while maximizing internal validity. Future studies could increase the ecological validity by using scenarios or stimuli that more closely resemble the symptoms or daily life. A further step could be the use of experience sampling method (ESM) or ecological momentary assessment (EMA) to offer valuable insights into the temporal dynamics of fear responses and their fluctuations over time in naturalistic environments (Verhagen et al., 2016), and help to capture individual variability in fear responses and conditioning processes (Vries et al., 2001). In addition, the strength of this meta-analysis lies in a comprehensive integration and summary of findings across studies, which, however, limits the resolution and inferences regarding the underlying processes and mechanisms that could not be integrated within this study. Nonetheless, it can guide the direction of future research by identifying which processes are consistently altered and which mechanisms require further elucidation. Finally, drawing from the insights garnered during the meta-analysis process, we have outlined the primary challenges encountered and formulated recommendations for future research (table 7).

Table 7

Challenges and recommendations

Challenge	Recommendation
Studies lack reporting relevant descriptive statistics or parameter estimates of the analyses.	- Include descriptive statistics (mean and standard deviations) for all outcome measurements. - Authors are encouraged to share or publish their data, in line with the open science movement (Branney et al., 2023; Renkewitz & Heene, 2019).
Hyperfocus on categorical diagnoses with limited use of questionnaires that allow for transdiagnostic analyses.	- Use measurements that facilitate dimensional analyses (e.g. Cross-D, Lebeau et al., 2012; 4DSQ, Terluin et al., 2006).
High degree of heterogeneity in the questionnaires used or in the assessed moderator variables (e.g., symptom severity, assessment of functioning).	- Utilization of standard questionnaires to promote integration across studies, (e.g. PHQ-9 (Kroenke et al., 2001), OCI-R (Foa et al., 2002; Gonner et al., 2007)) - Use and report standardization techniques like z-transformation to convert values into a common scale, facilitating the comparison and integration of results
Heterogeneity in terminology and research design when studying fear conditioning restricts comparability across studies.	- **Utilize the recommended shared terminology (Lonsdorf et al., 2017)** - Use standard design whenever possible **(Bach et al., 2023).**
Publication bias due to selective reporting of positive results in clinical trials	- Preregistration of study design and hypotheses - Justify sample size e.g., via a priori power analysis (Lakens, 2022) - **Transparent reporting and publication of all study outcomes, regardless of results.**
Different outcome measures reflect distinct learning systems (Ojala & Bach, 2020), and results often vary depending on the chosen outcome.	- To capture the complexity of fear conditioning, researchers should investigate a range of outcome measures to encompass the multiple facets of fear conditioning
Different outcome measurements require specific timings (Lonsdorf et al., 2017), which can lead to interference or alterations when recorded simultaneously	- Employ optimized designs for your dependent variables - If needed limit the number of outcomes - report results for all outcome measurements
Outcome measures differ in reliability and precision limiting the validity of results	- Report reliability or precision estimated for all outcome measures (i.e, questionnaires, ratings, physiological and neural outcomes)

5 Conclusions

Our meta-analysis integrated fear conditioning studies focusing on fear and safety learning in individuals with stress-relates and anxiety disorders and OCD. The results demonstrated heightened reactivity to safety cues during acquisition, extinction and extinction recall, supported by both verbal self-report and during acquisition also startle results. Additionally, individuals with stress- and anxiety-related disorders exhibited sustained exaggerated responses to threatening cues (CS+) during acquisition that sustain during extinction and extinction recall, as indicated by affect ratings. Effect sizes were generally small to medium. While alterations during acquisition were more pronounced (i.e. CS- responding) or even limited (CS+ responding) to anxiety and OCD, extinction effects were pronounced in PTSD. Given the considerable heterogeneity and signs of publication bias, as well as the predominance of results in self-report measures, caution is advised in drawing definitive conclusions. Further, the rather moderate support for alterations in fear and safety learning in PTSD, OCD and anxiety stands in contrast to the importance of these changes within many models of the onset and treatment of stress-related and anxiety disorders and underscores the necessity to expand our research to explore additional mechanisms and processes to understand and treat stress- and anxiety-related disorders. Overall, across studies, a small but consistent increase in responding to safety cues (CS-) is observed in stress- and anxiety-related disorders during acquisition, extinction and extinction recall, alongside a heightened response to threat cues (CS+) across these phases, suggesting resistance to extinction. These patterns underscore how stress- and anxiety-related disorders involve alterations in both threat and safety detection and fear regulation.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT (GPT-4; OpenAI, 2024) for language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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